

Oka properties of genus-one K3 surfaces and generalized Kummer surfaces

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Abstract

We prove that every complex K3 surface carrying a genus-one fibration is Oka: the class contains nonprojective torsors, and no section of the fibration is assumed. Two comparison theorems do the work. The first transfers the Oka property between complex manifolds whose complements of finite simple normal crossing divisors are Oka, provided that a diffeomorphism matches the divisors component by component and is biholomorphic near all of them; it imposes no Oka condition on the divisor strata and is available in every source dimension. The second exhibits a common holomorphic affine bundle over a genus-one torsor and over its Jacobian, and therefore shows that the two are Oka together. These tools reduce the problem to a single compact model, namely the K3 surface of discriminant three that arises as a cyclic triple cover of $\mathbb{P}^1 \times \mathbb{P}^1$ branched along six lines: its eleven genus-one fibrations separate tangent vectors, and a small deformation of it matches the whole exceptional neighbourhood and reaches Picard rank twenty. It follows that every complex K3 surface of Picard rank twenty is Oka, and so is every generalized Kummer K3 surface, that is, every minimal resolution of a quotient of a complex two-torus by a finite group of symplectic automorphisms; the case of inversion recovers all Kummer surfaces. We also show that every smooth complex Enriques surface is Oka, nodal ones included, and that the Oka locus in a non-isotrivial proper holomorphic family of K3 surfaces is a dense G_δ subset of the base.

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1 Introduction

The Oka property asks whether the classical approximation and interpolation theorems for holomorphic functions survive when the target is an arbitrary complex manifold. The question goes back to Oka and Grauert, but it received a geometric answer only with Gromov's notion of a dominating spray, which converts the existence of sufficiently many holomorphic deformations into the flexibility one is after [10, 8]. Forstnerič later gave this flexibility a purely approximation-theoretic form: a complex manifold X is Oka precisely when, for every $n \geq 1$ and every compact convex set $K \subset \mathbb{C}^n$, each map holomorphic near K can be approximated uniformly on K by entire maps $\mathbb{C}^n \rightarrow X$ [6]. Note that the source dimension is unrestricted, and this is exactly what separates the Oka property from approximation for maps whose source is an open Riemann surface.

For compact complex surfaces the earliest progress came from entire curves and dominating maps. Buzzard and Lu proved that Kummer surfaces, and more generally K3 surfaces admitting an elliptic fibration, are dominable by $\mathbb{C}^2$ [2, Section 4.2 and Theorem 4.7]; Forstnerič and Lárusson then showed that every Kummer surface is strongly dominable and that the complement of its sixteen exceptional curves is Oka [9, Corollary 3 and Lemma 7];

and Alarcón and Forstnerič established the Oka-1 property for Kummer surfaces and elliptic K3 surfaces alike [1, Proposition 8.4 and Corollary 8.6]. Oka-1, however, controls only maps whose source is an open Riemann surface, whereas the full Oka property must also handle maps from Stein manifolds of arbitrary dimension; the compact surfaces themselves were therefore left open.

More recently, Xie and Zhao produced Oka K3 surfaces from smooth hypersurfaces of multidegree $(2, 2, 2)$ in $(\mathbb{P}^1)^3$, and proved that the Oka locus is a dense G_δ subset of every local Kuranishi family [21, Theorems A and B]; their methods also give projective and nonprojective Oka Kummer surfaces, and they cover unnodal Enriques surfaces [21, Theorems C and E]. Those constructions provide compact examples together with deformation statements. Here we take a different route: instead of exhibiting one model at a time, we compare whole classes of K3 surfaces with models already known to be Oka, in a manner that leaves intact the geometry responsible for approximation.

The difficulty is already visible on a Kummer surface. Its complement of exceptional curves is Oka, but approximation in that complement tells us nothing about the compact surface, because a map meeting one of the curves transversely continues to meet it after any sufficiently small perturbation. One must instead compare sections of the two manifolds along the curves themselves, and that is what our first comparison theorem achieves (Theorem 2.1): if two complex manifolds have Oka complements of finite simple normal crossing divisors, and if a diffeomorphism identifies the divisors component by component while being biholomorphic near all of them, then the Oka property transfers in both directions. Combined with explicit compact Oka models and with a comparison between a genus-one torsor and its Jacobian, this yields the main theorem of the paper.

Throughout, a genus-one fibration means a surjective holomorphic map from a smooth compact K3 surface to $\mathbb{P}^1$ whose fibres are connected and whose general fibre is smooth of genus one. We identify fibrations that differ by an automorphism of the base. We insist on no section, and we keep singular fibres with all their components and multiplicities, since the arguments below have to survive both the absence of a section and the degenerations of the fibres.

Theorem 1.1. *Every complex K3 surface admitting a genus-one fibration is Oka.*

Idea of the proof

We prove the theorem in three steps, and it is worth saying at the outset where the difficulty lies: not in producing sprays on a compact surface, but in keeping them under control across the exceptional curves.

The first step removes the group law from the picture. A genus-one K3 surface need not be its own Jacobian, but the two are related by a common holomorphic affine bundle: a single complex manifold is the total space of a holomorphic affine bundle over the torsor and also of one over its Jacobian, and the Oka property passes through affine bundles in both directions. It is therefore enough to prove the theorem for Jacobians, the only price being that the bundle comparison must survive singular fibres. It does, because the Hamiltonian action of the fibration is complete on every fibre, singular ones included.

The second step provides the compact model. A genus-one fibration on a K3 surface carries a holomorphic motion along its fibres, obtained by contracting the holomorphic two-form against the differential of the fibration, and compactness of the fibres makes this motion complete for all complex times, so that it integrates to an additive spray. One spray never dominates, but several fibrations can: when their Hamiltonian directions span the tangent space at every point, the surface is subelliptic and hence Oka. This mechanism

is due to Xie and Zhao [21, Section 3], and what is needed here is an explicit surface for which it can be checked by hand; a discriminant-three model given by a triple cover of $\mathbb{P}^1 \times \mathbb{P}^1$ serves that purpose.

The hard part is the third step, and it is the reason for the normal-crossing transfer. Exceptional curves are precisely where flexibility is most difficult to control, since a spray that is complete on the complement need not extend across them. Our transfer avoids the issue by keeping an entire neighbourhood of the divisors holomorphically identified, while the Oka complements gives the deformations away from them; the point of following, separately for each divisor component, how a map meets that component is that the deformations can then be joined across the curves. The transfer assumes nothing about the divisor strata themselves being Oka, and it works in every source and target dimension. Applied to the marked Inose companions it produces compact Oka models of Picard rank twenty.

Finally, two genus-one fibrations whose fibre classes meet in at most three points have an Oka complement of their common contracted curves, and matching the whole exceptional neighbourhood of such a surface with a Picard-rank-twenty model makes the compact surface itself Oka. For a prescribed Jacobian, a lattice construction provides a surface of this kind whose Jacobian is the one prescribed, so that the affine-bundle comparison first gives the Oka property of that Jacobian and then carries it back to the original torsor.

Projectivity enters only through the auxiliary models, where it provides the contraction and the ample class needed for the deformation comparison and the polarization used to reconstruct the Jacobian. The affine-bundle comparison returns the Oka property to the original analytic torsor, which need not be projective.

Further consequences

By a generalized Kummer surface we mean the minimal resolution of a quotient A/G which happens to be a K3 surface, where A is a complex two-torus and G is a finite group of holomorphic automorphisms preserving a nonzero holomorphic two-form. The automorphisms are allowed to have translation parts, and they need not share a fixed point; the term is used here for surfaces only, and not for the higher-dimensional generalized Kummer varieties.

Theorem 1.2. *Every generalized Kummer surface is Oka.*

The proof is again a comparison. An equivariant construction produces a projective generalized Kummer surface of Picard rank at least seventeen, and a projective K3 surface of Picard rank at least five carries a genus-one fibration, so Theorem 1.1 applies to that model and the normal-crossing transfer brings the conclusion back. Section 9 also exhibits nonprojective examples of order three, which have neither a genus-one fibration nor an ordinary Kummer structure.

Corollary 1.3. *Every complex Kummer surface is Oka.*

This is the case $G = \{\pm 1\}$ of Theorem 1.2. The question is recorded in [5, Section 6.2] and [8, Section 7.2], and the corollary extends the compact examples of [21] to Kummer surfaces of arbitrary complex two-tori, projective or not.

Corollary 1.4. *Every smooth complex Enriques surface is Oka, including every nodal Enriques surface.*

Recall that a nodal Enriques surface is a smooth Enriques surface containing a (-2) -curve. Its canonical K3 cover is unramified, and it stays unramified along those curves, so applying the genus-one theorem to the cover extends the unnodal result of [21] to all smooth complex Enriques surfaces.

The genus-one theorem is also strong enough to detect Oka fibres inside an arbitrary nonconstant family of K3 surfaces.

Theorem 1.5. *If $\mathcal{X} \rightarrow B$ is a non-isotrivial proper holomorphic submersion with K3 fibres over a connected complex manifold, then its Oka locus is a dense G_δ subset of B .*

Here non-isotrivial means that the complex structures are not locally constant up to biholomorphism, and density is understood in the complex-analytic topology. The statement does not follow from the Kuranishi-family result quoted above, because the base may trace only a lower-dimensional part of the period domain, so that the period cannot be varied freely. What the proof does instead is to find, densely along every nonconstant period curve, fibres carrying an integral isotropic Néron–Severi class; Theorem 1.1 makes those fibres Oka, and Lárusson’s deformation theorem upgrades density to a dense G_δ statement [14, Corollary 8].

The paper follows this order. Section 2 proves the normal-crossing transfer and Section 3 the completeness of the flows along a fibration; Section 4 gives the torsor–Jacobian reduction. The compact models are built in Sections 5 and 6, after which Sections 7 and 8 prove the genus-one theorem through the small-intersection and Jacobian constructions. The final three sections treat torus quotients, Enriques surfaces and non-isotrivial families, and Appendix A records the geometric comparison between arbitrary Kummer models.

2 Transfer across simple normal crossing divisors

2.1 The transfer statement and relative sections

We begin with the tool that lets us move the Oka property from one compact surface to another. A finite simple normal crossing divisor is a reduced divisor $E = E_1 + \cdots + E_r$ whose irreducible components are smooth and are locally given by distinct coordinate hyperplanes; here and below we also write E for its support, and we label the components whenever two such divisors are to be compared.

Theorem 2.1. *Let X and Y be connected complex manifolds, and let $E = E_1 + \cdots + E_r$ and $F = F_1 + \cdots + F_r$ be finite simple normal crossing divisors. Suppose that $X \setminus E$ and $Y \setminus F$ are Oka. If there is a smooth diffeomorphism $h : X \rightarrow Y$ satisfying $h(E_i) = F_i$ for every i , which is biholomorphic on a neighbourhood of E , then X is Oka if and only if Y is Oka.*

The theorem is stated for an arbitrary pair of divisors, but the case we have in mind is that of the exceptional divisors with D - and E -type dual graphs that occur on the surfaces below. What makes the proof work is that a separate normal ratio is kept for each divisor component, and that this bookkeeping is maintained at the crossings as well.

Two conventions are in force throughout. A holomorphic submersion is understood locally as a holomorphic projection, and when we call a map on a reduced analytic subset holomorphic we refer to its reduced complex-space structure.

Lemma 2.2. *Let S be a Stein manifold, let $D \subset S$ be a closed reduced complex subvariety, and let $q : Q \rightarrow S$ be a holomorphic submersion. Suppose that q restricts over $S \setminus D$ to a holomorphic fibre bundle with Oka fibre. Let $C \subset S$ be compact and $\mathcal{O}(S)$ -convex.*

Every continuous section s which is holomorphic near C and whose restriction to D is holomorphic is homotopic, through continuous sections fixed on D , to a holomorphic section arbitrarily close to s on C . The homotopy can also be kept arbitrarily close to s on C .

Proof. We construct the homotopy by an exhaustion of S , alternating the two results of Forstnerič on sections of stratified fibre bundles that we quoted above [7, Propositions 4.1 and 5.1]; throughout, the stratification is the pair $M_0 = S$, $M_1 = D$. We choose a complete distance on Q and an exhaustion

$$C = C_0 \subset C_1 \subset C_2 \subset \cdots$$

by compact $\mathcal{O}(S)$ -convex sets, with $C_j \subset \text{int } C_{j+1}$ for $j \geq 1$. Note that $C_j \cap D$ is $\mathcal{O}(D)$ -convex: restrictions of functions separating a point of $D \setminus C_j$ from C_j also separate it from $C_j \cap D$.

Suppose that the section we have is holomorphic near C_j and on D . By [7, Proposition 4.1], there is a holomorphic section on a neighbourhood of $C_j \cup D$, equal to the current section on D and arbitrarily close to it on C_j . The new and old sections can be joined by short paths in the fibres near $C_j \cup (D \cap C_{j+2})$: the two are close near C_j and agree on D , so continuity permits a sufficiently small neighbourhood of $D \cap C_{j+2}$ on which these paths are defined. A cutoff in the base then replaces the old section by the new one near this set, while preserving it outside a slightly larger neighbourhood. The resulting homotopy is fixed on D , and its displacement on C_j is arbitrarily small.

It remains to make the section holomorphic near C_{j+2} . For this we apply [7, Proposition 5.1] with $K = C_j$, $L = C_{j+2}$, $M_0 = S$, and $M_1 = D$. It provides a section homotopy fixed on $D \cap C_{j+2}$, arbitrarily small on C_j , whose final section is holomorphic near C_{j+2} . Cut off its time parameter by a function equal to one near C_{j+1} and supported in $\text{int } C_{j+2}$; this makes the homotopy fixed on all of D and leaves its final section holomorphic near C_{j+1} .

We choose the total error of the j th step smaller than $2^{-j-2}\epsilon$. Concatenation on intervals tending to 1 gives a homotopy with a locally uniform limit at time 1, and completeness ensures that the limit exists. Since the successive sections are eventually holomorphic on a neighbourhood of each C_j , that locally uniform limit is again a holomorphic section. The homotopy is fixed on D and stays within ϵ of the initial section on C . The argument also applies when C is empty. $\square$

2.2 Transversality and simultaneous comparison

The comparison below has to be carried out for all components at once, so we record the relevant notion of transversality. For a finite simple normal crossing divisor $E = \sum_i E_i$, transversality to its strata means transversality to every nonempty intersection of its components; at a point of an intersection, this is equivalent to linear independence of the pulled-back local defining differentials.

Lemma 2.3. *Let φ be holomorphic on a neighbourhood of a compact convex set $K \subset \mathbb{C}^n$, with values in a complex manifold X . For every finite simple normal crossing divisor $E \subset X$, the map φ can be approximated uniformly on K by maps holomorphic on a smaller convex neighbourhood of K and transverse to all strata of E .*

Proof. We choose a convex Stein neighbourhood U of K on which φ is holomorphic. The graph of φ has a Stein neighbourhood in $U \times X$ by [18], so Cartan's theorem A, applied to the vertical tangent bundle, provides finitely many holomorphic vertical vector fields spanning that bundle along the graph over a compact neighbourhood of K . Their local flows produce a holomorphic family φ_t , defined on a fixed smaller neighbourhood for all

sufficiently small parameters, whose parameter differential is surjective at $t = 0$; after decreasing the source and parameter neighbourhoods, the total map is a submersion.

Now parametric transversality can be applied to every nonempty intersection of the components of E . There are finitely many such intersections, so arbitrarily small parameters make φ_t transverse to all of them, and restriction to a convex neighbourhood of K proves the assertion. $\square$

Lemma 2.4. *Let $U \subset \mathbb{C}^n$ be Stein, and let $\phi_1, \dots, \phi_r \in \mathcal{O}(U)$. Assume that, at each point $z \in U$, the differentials $d\phi_i(z)$ for which $\phi_i(z) = 0$ are linearly independent. Suppose that $\psi_{\nu,i} \in \mathcal{O}(U)$ and that $\psi_{\nu,i} \rightarrow \phi_i$ locally uniformly on U . Given a compact set $L \Subset U$, there are fixed smaller neighbourhoods of L and holomorphic maps β_ν and functions $\ell_{\nu,i}$ on them, for all sufficiently large ν , such that*

$$\psi_{\nu,i} \circ \beta_\nu = e^{\ell_{\nu,i}} \phi_i, \quad 1 \leq i \leq r. \quad (1)$$

The maps β_ν are biholomorphisms onto their images, which contain a fixed smaller neighbourhood of L . On fixed smaller neighbourhoods, β_ν and their inverses converge to the identity, and $\ell_{\nu,i} \rightarrow 0$.

Proof. We look for the biholomorphisms β_ν as time-one maps of a flow that straightens the functions $\psi_{\nu,i}$ against the ϕ_i . We consider the holomorphic bundle map

$$\mathcal{D}_\phi : TU \oplus \mathcal{O}_U^r \longrightarrow \mathcal{O}_U^r, \quad \mathcal{D}_\phi(V, c) = (d\phi_i(V) - c_i \phi_i)_{i=1}^r. \quad (2)$$

This bundle map is surjective. For each index with $\phi_i \neq 0$, the variable c_i controls the corresponding coordinate, while the vector V controls the remaining coordinates by the assumed independence of the differentials belonging to the vanishing functions. Cartan's theorem B therefore provides a holomorphic right inverse R_0 .

Let $\delta_{\nu,i} = \psi_{\nu,i} - \phi_i$ and $\phi_{i,t} = \phi_i + t\delta_{\nu,i}$. We fix nested compact neighbourhoods of L ; by Cauchy estimates the convergence is C^1 on the smaller ones. For t in a bounded complex neighbourhood of $[0, 1]$, consider the family

$$R_t = R_0(I + (\mathcal{D}_{\phi_t} - \mathcal{D}_\phi)R_0)^{-1}$$

The inverse exists for large ν , uniformly in this neighbourhood of $[0, 1]$, so R_t is holomorphic and uniformly bounded on a fixed smaller source neighbourhood. It is a right inverse of $\mathcal{D}_{\phi_t}$.

The vector field and coefficient tuple $(V_t, c_t) = -R_t \delta_\nu$ satisfy

$$\partial_t \phi_{i,t} + d\phi_{i,t}(V_t) = c_{i,t} \phi_{i,t}, \quad \|V_t\| + \|c_t\| \longrightarrow 0$$

uniformly on the fixed compact neighbourhoods. Since the field is uniformly small, the holomorphic flow β_t of V_t , beginning at the identity, exists for $0 \leq t \leq 1$ on a fixed smaller neighbourhood, and its inverse exists on a further fixed smaller neighbourhood; both remain in the domain of the coefficients. Along the flow,

$$\phi_{i,t}(\beta_t(z)) = \exp\left(\int_0^t c_{i,s}(\beta_s(z)) ds\right) \phi_i(z).$$

Taking $t = 1$ gives (1), and the uniform smallness of the vector field and of the coefficients also gives the stated convergence on the fixed domains. $\square$

2.3 Interpolation near a holomorphic section

The next statement is what allows us to prescribe the behaviour of a section on the divisor while approximating it elsewhere; it is applied later to sections of an auxiliary submersion rather than to maps into the surface itself.

Lemma 2.5. *Let $q : Q \rightarrow U$ be a holomorphic submersion over a Stein manifold, and let $D \subset U$ be a closed reduced complex subvariety. Let σ be a holomorphic section, and let τ be a continuous section homotopic to σ through sections fixed on D . Suppose that τ_ν are continuous sections, holomorphic near D , which converge to τ uniformly on compact subsets. For every compact $L \Subset U$, after restriction to a fixed smaller neighbourhood of L and passage to a subsequence, there are holomorphic sections $\sigma_\nu \rightarrow \sigma$ and section homotopies from τ_ν to σ_ν fixed on D .*

Proof. We choose nested relatively compact Stein neighbourhoods of L . Along σ , holomorphic fibre coordinates identify a neighbourhood of the zero section of the bundle

$$V = \sigma^*(\ker dq)$$

with a neighbourhood of σ in Q , preserving the projection. To obtain such coordinates, take a Stein neighbourhood of the section, choose finitely many vertical holomorphic vector fields spanning along a compact neighbourhood of L , and compose their small flows; Cartan's theorem B splits the surjective parameter differential along the section, and restricting the flow map to a complementary subbundle and applying the inverse function theorem produces the asserted coordinates. This is the fibre-preserving version of the construction in [7, Section 3], and the coordinate domains can be made fibrewise convex after decreasing them.

Since $\tau|_D = \sigma|_D$, local uniform convergence places the restrictions $\tau_\nu|_D$ in these coordinates on an intermediate relatively compact Stein neighbourhood U_2 , for all sufficiently large ν ; they correspond to holomorphic sections v_ν of $V|_{D \cap U_2}$ tending to zero. By Cartan's theorem B the restriction map

$$\Gamma(U_2, V) \longrightarrow \Gamma(D \cap U_2, V|_D) \quad (3)$$

is a continuous surjection of Fréchet spaces, so the open mapping theorem implies that, after passage to a subsequence, v_ν admits holomorphic extensions tending to zero on the closure of a fixed smaller neighbourhood $U_1 \Subset U_2$. Applying the fibre coordinates to these extensions produces holomorphic sections σ_ν on U_1 with

$$\sigma_\nu \longrightarrow \sigma, \quad \sigma_\nu|_D = \tau_\nu|_D. \quad (4)$$

On a still smaller fixed neighbourhood, join τ_ν to τ by short paths in the fibres, follow the given homotopy from τ to σ , and then join σ to σ_ν in the holomorphic fibre coordinates. Over D the middle part is constant and the other two parts are uniformly small, so by compactness there is a neighbourhood of the relevant part of D on which the entire concatenated path lies in a fibrewise convex coordinate neighbourhood of σ . We write its coordinates there as $w_t(z)$.

We choose a cutoff λ supported in that neighbourhood and equal to one on D over the closure of a fixed smaller neighbourhood U_0 of L , and replace w_t by

$$(1 - \lambda)w_t + \lambda((1 - t)w_0 + tw_1). \quad (5)$$

Convexity makes this a valid path, and both endpoints are unchanged. Equation (4) implies $w_0 = w_1$ on D , so the modified path is constant there. Restriction to U_0 completes the proof. $\square$

2.4 The auxiliary submersion

Everything so far has been prepared for one construction, in which a map into one of our two surfaces is encoded by a section of a submersion whose fibre over the complement of a divisor is a product. Let S be a complex manifold, and for each i let $L_i \rightarrow S$ be a holomorphic line bundle with a holomorphic section d_i . Let $M_i = \mathcal{O}_X(E_i)$, with canonical section e_i . In the product over $S \times X$ of the bundles of invertible homomorphisms $L_i \rightarrow M_i$, consider

$$Q_X = \{(z, x, \eta_1, \dots, \eta_r) : \eta_i : (L_i)_z \xrightarrow{\sim} (M_i)_x, \\ e_i(x) = \eta_i(d_i(z)) \text{ for every } i\}. \quad (6)$$

Let $q_X : Q_X \rightarrow S$ be the projection. Construct Q_Y in the same way from $\mathcal{O}_Y(F_i)$ and its canonical section, keeping the source data (L_i, d_i) fixed.

Lemma 2.6. *The space Q_X is a complex manifold and q_X is a holomorphic submersion. If D is the reduced union of the zero sets of the d_i , then*

$$Q_X|_{S \setminus D} \simeq (S \setminus D) \times (X \setminus E) \quad (7)$$

over $S \setminus D$. A diffeomorphism h as in Theorem 2.1 induces a diffeomorphism $\hat{h} : Q_X \rightarrow Q_Y$ over S , biholomorphic on a neighbourhood of $q_X^{-1}(D)$.

Proof. At a point of Q_X , let $I = \{i : d_i(z) = 0\}$. The equations in (6) imply that these are exactly the indices of the components of E containing x , and in local frames the simple normal crossing condition makes their canonical sections coordinate functions y_i on X . The equations for $i \in I$ are

$$y_i = d_i(z)\eta_i, \quad \eta_i \neq 0.$$

They eliminate the coordinates y_i , while for $j \notin I$ the equation uniquely determines η_j , since $d_j(z) \neq 0$; the remaining coordinates include all base coordinates freely. This proves smoothness and submersivity, and outside D all ratios are uniquely determined, which proves (7).

There are canonical smooth complex line-bundle isomorphisms

$$M_i \simeq h^* \mathcal{O}_Y(F_i) \quad (8)$$

which preserve the canonical sections. Near E they are induced holomorphically by h , while outside E_i preservation of the nonzero canonical section determines them uniquely; the two constructions agree on the overlap and hence glue smoothly. Composition with the η_i gives $\hat{h}$, whose inverse is induced by h^{-1} . Over D the point x lies in E , so the induced map is holomorphic on a neighbourhood of $q_X^{-1}(D)$. $\square$

2.5 Proof of the transfer theorem

The proof runs the auxiliary submersions through three stages. We first deform the local section over the Oka manifold Y ; next we approximate its underlying map by entire maps and align all the pulled-back divisor components simultaneously; and finally a relative section homotopy produces local holomorphic data on the resulting global auxiliary submersion, at which point the Oka property of $X \setminus E$ gives an entire map into X .

Proof of Theorem 2.1. Assume that Y is Oka, and let $K \subset \mathbb{C}^n$ be compact and convex, let φ be holomorphic near K with values in X , and fix $\epsilon > 0$. By Lemma 2.3, after an initial

perturbation of size less than $\epsilon/3$, the map is transverse to all strata of E on a convex Stein neighbourhood U of K ; we retain the notation φ for the perturbed map.

Let $L_i = \varphi^* M_i$ and $d_i = \varphi^* e_i$. The reduced union D of their zero sets is a simple normal crossing divisor in U . The map φ determines the holomorphic section $\sigma : U \rightarrow Q_X$ whose homomorphisms are the tautological identities, and the section $\hat{h}\sigma$ of Q_Y is continuous and holomorphic near D . Lemmas 2.2 and 2.6, with empty approximation compact, provide a homotopy of sections s_t of Q_Y , fixed on D , such that

$$s_0 = \hat{h}\sigma, \quad s_1 \text{ is holomorphic on } U. \quad (9)$$

We write $\psi : U \rightarrow Y$ for the underlying map of s_1 . Its line homomorphisms identify (L_i, d_i) holomorphically with the pullbacks of the canonical pairs for F_i , so that

$$\psi^* F_i = \varphi^* E_i \quad (10)$$

as effective divisors; moreover, since the nonzero normal ratios preserve the independence of the local defining differentials, the transversality of φ implies that of ψ to all strata of F . The section $\tau = \hat{h}^{-1}s_1$ is joined to σ by a section homotopy fixed on D , namely the reverse of (9) transported by $\hat{h}^{-1}$, and therefore $\tau|_D = \sigma|_D$.

Since Y is Oka and U is convex, there are entire maps $\psi_\nu : \mathbb{C}^n \rightarrow Y$ converging to ψ uniformly on compact subsets of U . We choose a point of $U \setminus D$: for large ν , its image under ψ_ν lies outside F , so none of the canonical pullback sections below is identically zero. The global source pairs on $\mathbb{C}^n$ are

$$L_{i,\nu} = \psi_\nu^* \mathcal{O}_Y(F_i), \quad d_{i,\nu} = \psi_\nu^* e_{F_i},$$

and D_ν denotes the reduced union of their zero sets. These pairs determine global auxiliary submersions $Q_{X,\nu}$ and $Q_{Y,\nu}$. Transporting the tautological holomorphic section of $Q_{Y,\nu}$ by $\hat{h}^{-1}$ gives a continuous section A_ν of $Q_{X,\nu}$ which is holomorphic near D_ν .

We compare these submersions near K . We choose fixed nested convex neighbourhoods with compact inclusions in U ; each L_i is trivial there. We use the line homomorphisms of s_1 to transfer a frame of L_i to a frame of $\mathcal{O}_Y(F_i)$ along the graph of ψ . A Stein neighbourhood of that graph exists by [18]. Cartan's theorem B extends the frame to this neighbourhood; its nonzero locus is an open neighbourhood of the graph, and taking the intersection of these loci for all i provides simultaneous frames on one common neighbourhood.

In these frames, let $\phi_i(z)$ and $\psi_{\nu,i}(z)$ be the coefficients of the canonical sections along ψ and ψ_ν ; the coefficient ϕ_i is also that of d_i in the chosen frame of L_i . These holomorphic coefficient functions satisfy $\psi_{\nu,i} \rightarrow \phi_i$ locally uniformly, so Lemma 2.4 provides β_ν and $\ell_{\nu,i}$ satisfying (1).

Multiplication by $e^{\ell_{\nu,i}}$ in the common frames gives an isomorphism

$$\theta_{i,\nu} : L_i \longrightarrow \beta_\nu^* L_{i,\nu}, \quad \theta_{i,\nu}(d_i) = \beta_\nu^* d_{i,\nu}.$$

It induces an isomorphism C_ν from the fixed local space Q_X to $Q_{X,\nu}$ over β_ν , by replacing η_i with $\eta_i \circ \theta_{i,\nu}^{-1}$; in particular,

$$\beta_\nu^{-1}(D_\nu) = D \quad (11)$$

on the comparison neighbourhood.

We pull A_ν back by C_ν and denote the resulting local section of Q_X by τ_ν . Before applying $\hat{h}^{-1}$, its underlying map is $\psi_\nu \circ \beta_\nu$ and its homomorphisms have coefficients $e^{\ell_{\nu,i}}$ in the common frames, so they converge to the corresponding data of s_1 . Consequently,

$$\tau_\nu \longrightarrow \tau \quad (12)$$

uniformly on fixed smaller compact neighbourhoods, including points of D , and each τ_ν is holomorphic near D .

Now apply Lemma 2.5. After decreasing the fixed neighbourhood and passing to a subsequence, it provides holomorphic sections $\sigma_\nu \rightarrow \sigma$ and local section homotopies $J_{\nu,t}$ from τ_ν to σ_ν , fixed on D . We choose a sufficiently large index so that the comparison image contains a fixed neighbourhood of K and $\beta_\nu^{-1}(K)$ lies in the domain of these homotopies, transport the homotopy to $Q_{X,\nu}$ by C_ν , and cut off its time parameter by a function equal to one near K and compactly supported in that comparison image. Together with A_ν outside the support, this gives a global continuous section H_ν of $Q_{X,\nu}$.

Near K , the section H_ν is the holomorphic transport of σ_ν . The divisor alignment (11) and the fact that $J_{\nu,t}$ is fixed on D give

$$H_\nu|_{D_\nu} = A_\nu|_{D_\nu}. \quad (13)$$

Its restriction to the reduced analytic space D_ν is therefore holomorphic.

Lemma 2.2 now applies to $Q_{X,\nu} \rightarrow \mathbb{C}^n$, by (7) and the Oka property of $X \setminus E$. It produces a global holomorphic section arbitrarily close to H_ν on K , whose projection to X is an entire map. Before the final approximation, the projected map near K is the underlying map of σ_ν composed with β_ν^{-1} , so the convergence of both factors gives uniform convergence to φ on K . We choose the index and the final approximation so that their errors are each less than $\epsilon/3$; together with the initial perturbation, this proves convex approximation for X .

The inverse diffeomorphism satisfies the same hypotheses with X and Y exchanged, which proves the equivalence. $\square$

2.6 Resolved involution quotients

The transfer theorem is used in this paper almost always in the following special shape, where the divisors are the exceptional curves of a resolved involution quotient.

Corollary 2.7. *For $i = 0, 1$, let S_i be a connected compact complex surface with an involution ν_i whose fixed set P_i is finite and whose tangent action there is $-\text{id}$. Put*

$$Y_i = \text{Bl}_{P_i} S_i, \quad X_i = Y_i / \nu_i.$$

Suppose $S_i \setminus P_i$ is Oka for both indices and an equivariant diffeomorphism $h : S_1 \rightarrow S_0$ is biholomorphic near P_1 . Then X_0 is Oka if and only if X_1 is Oka.

Proof. Holomorphic linearization at a fixed point gives the action $\xi \mapsto -\xi$ on $\mathbb{C}^2$. In a blowup chart $\xi = (y, yv)$ it becomes

$$(y, v) \mapsto (-y, v), \quad (y, v) \mapsto (y^2, v)$$

for the action and quotient map. Hence X_i is smooth and its exceptional divisor B_i is a disjoint union of rational (-2) -curves. The equivariant map h lifts to the blowups: near the centers this is the holomorphic lift, while away from them it is defined by the blowdown isomorphisms, and these lifts agree on overlaps. They descend to a smooth diffeomorphism $X_1 \rightarrow X_0$ biholomorphic near the whole B_1 .

The complement $X_i \setminus B_i$ is the unramified quotient of $S_i \setminus P_i$ by its free involution, so it is Oka by covering invariance [6, Theorem 1.4], and Theorem 2.1 applies in either direction. $\square$

2.7 Localizing an equivariant isotopy

One further construction is needed, which lets us make a smooth comparison linear near finitely many distinguished points.

Lemma 2.8. *Let $H \subset \mathrm{GL}(\mathbb{R}^4)$ be a finite group, and let $(M_t)_{0 \leq t \leq 1}$ be a smooth path in $\mathrm{GL}(\mathbb{R}^4)$ with $M_0 = \mathrm{id}$. Suppose that every M_t commutes with H . For every H -invariant ball B centred at the origin, there is an H -equivariant smooth diffeomorphism of $\mathbb{R}^4$ which is the identity outside B and agrees with M_1 near the origin.*

Proof. We choose an H -invariant inner product and a smooth H -invariant cutoff χ compactly supported in B , with $\chi = 1$ on a ball of radius $\rho > 0$ about the origin. The time-dependent vector field

$$V_t(x) = \chi(x) \dot{M}_t M_t^{-1} x \quad (14)$$

is H -equivariant and has compact support independent of t . Its flow F_t therefore exists for $0 \leq t \leq 1$, consists of smooth diffeomorphisms, and is the identity outside B ; uniqueness makes the flow H -equivariant as well.

Now choose $\delta > 0$ such that

$$\delta \sup_{0 \leq t \leq 1} \|M_t\| < \rho.$$

For $\|x\| < \delta$, the path $M_t x$ remains where $\chi = 1$ and solves the flow equation with initial value x , so uniqueness gives $F_t(x) = M_t x$ there. The time-one map F_1 therefore has the required properties. $\square$

3 Complete flows along a fibration

A genus-one fibration on a K3 surface comes with a canonical family of holomorphic motions, and it is these that will produce the sprays used later. Let S be a K3 surface, ω a nonzero holomorphic two-form, and $f : S \rightarrow \mathbb{P}^1$ a genus-one fibration. Symplectic contraction gives

$$A_f = \omega^\sharp \circ df^* : f^* T^* \mathbb{P}^1 \longrightarrow T_S.$$

Lemma 3.1. *The map A_f integrates to a holomorphic additive action of $T^* \mathbb{P}^1$ along f , on the complete surface and all its singular fibres.*

Proof. Skew-symmetry gives $df \circ A_f = 0$. On $f^* T^* \mathbb{P}^1$, the vector field $(A_f(x, v), 0)$ keeps the value $f(x)$ and the cotangent parameter v fixed, so its trajectory stays in the compact fibre $f^{-1}(f(x)) \times \{v\}$; holomorphic ODE continuation therefore gives a flow for every complex time, and its time-one map is holomorphic in all variables.

In a local frame, the vector fields have the form $(a \circ f)V$. They commute, because the vertical field V annihilates their coefficients; linearity in v gives additivity, and negation of the parameter gives the inverse. Note that this argument also applies where the vector field vanishes, on nonreduced fibres, and at intersections of fibre components. It is the construction of [21, Proposition 3.2]. $\square$

Recall that a spray associated with a holomorphic map $f : M \rightarrow B$ and a vector bundle $E \rightarrow B$ is a holomorphic map $s : f^* E \rightarrow M$ satisfying

$$s(x, 0) = x, \quad f(s(x, e)) = f(x), \quad s(s(x, e), e') = s(x, e + e'). \quad (15)$$

The last identity is the additive property. For a local section a of E , the map $\tau_a(x) = s(x, a(f(x)))$ is a biholomorphism on its base preimage, with inverse τ_{-a} .

Lemma 3.2. *Let a complex manifold M have finitely many additive sprays as in (15), and let U_0 be the locus where their vertical differentials span T_M . If each point can be carried into U_0 by a finite composition of the local maps τ_a , their inverses, and holomorphic automorphisms of M , then M is Oka.*

Proof. We follow the induction of Xie and Zhao [21, Proposition 3.6]. Kusakabe's criterion [12, Corollary 4.1] reduces the conclusion to the following assertion: for each holomorphic $h : S \rightarrow M$ from a Stein manifold and each $s_0 \in S$, finitely many holomorphic sprays over h have parameter differentials spanning $T_{h(s_0)}M$. We prove this assertion by induction on the length of a composition that carries $h(s_0)$ into U_0 , treating first the two kinds of step such a composition may contain.

For an automorphism step, apply the induction to the automorphism composed with h , and then compose the resulting sprays with its inverse. If instead $h(s_0)$ already lies in U_0 , then Cartan's theorem A provides finitely many global sections of each $(f \circ h)^*E$ spanning its fibre at s_0 , and inserting their linear combinations into the additive sprays proves the assertion at that point.

It remains to handle a translation step. Suppose the first map of the composition is τ_a for one spray $s : f^*E \rightarrow M$. A global section p of $(f \circ h)^*E$ can be chosen with $p(s_0) = a(f(h(s_0)))$, and then

$$h_1(s') = s(h(s'), p(s'))$$

takes the required translated value at s_0 . By induction there are sprays $H_k : S \times \mathbb{C}^{N_k} \rightarrow M$ over h_1 whose parameter differentials span at s_0 . Since $f \circ h_1 = f \circ h$, Cartan's theorem B extends $-p$ from the zero section to a global section q_k of $(f \circ H_k)^*E$, and the maps

$$\hat{H}_k(s', t) = s(H_k(s', t), q_k(s', t))$$

are sprays over h by (15).

Let $\pi : f^*E \rightarrow M$ be the projection and let $\xi = (h_1(s_0), -p(s_0))$. Lifting each H_k by its section q_k gives $\tilde{H}_k$ with

$$\ker d\pi_\xi + \sum_k \operatorname{im} d_t \tilde{H}_k(s_0, 0) = T_\xi(f^*E).$$

The differential ds_ξ is surjective: along the local section $-a$, it is the differential of the inverse translation, and by additivity its image on $\ker d\pi_\xi$ is exactly the original spray direction at $h(s_0)$. Applying ds_ξ to the displayed tangent-space equality therefore shows that the parameter differentials of the $\tilde{H}_k$, together with those of the sprays over h given by Cartan's theorem A for the original direction, span the target tangent space, which completes the induction. Note finally that every constructed spray has a full affine parameter domain, as required by Kusakabe's criterion. $\square$

Let us record how the sprays of Lemma 3.1 can be made to avoid a prescribed finite set. For a finite set $P \subset S$, choose a reduced divisor $D_f \subset \mathbb{P}^1$ containing $f(P)$, and restrict the action through

$$T^*\mathbb{P}^1(-D_f) \longrightarrow T^*\mathbb{P}^1 \tag{16}$$

to fix the fibres over D_f pointwise. Every trajectory over the complement of D_f then stays in a fibre disjoint from P , so the restricted action defines an entire additive spray on $S \setminus P$, with the original Hamiltonian direction wherever $f(x) \notin D_f$. We call these the protected sprays, and say that the fibres over D_f are protected.

4 An affine-bundle equivalence for additive twists

The next step reduces the genus-one theorem to the case of Jacobian elliptic K3 surfaces. The mechanism is a comparison of two complex manifolds through a common affine bundle: we first show that a manifold and its additive twist over the same base have the same Oka status, and then apply the complete fibrewise action of Lemma 3.1 to analytic torsors.

Let $\pi : Y \rightarrow B$ be a holomorphic map between complex manifolds, let $E \rightarrow B$ be a holomorphic vector bundle, and suppose that

$$s : Y \times_B E \longrightarrow Y$$

is a holomorphic action of its additive group bundle, so that $\pi(s(y, v)) = \pi(y)$, $s(y, 0) = y$, and $s(s(y, v), w) = s(y, v + w)$. Twisting such an action along a cocycle is the operation we want to control. We choose an additive Čech cocycle a_{ij} for E over a cover $\{U_i\}$ of B , and let $P \rightarrow B$ be the corresponding E -torsor, with local coordinates satisfying $p_j = p_i + a_{ij}$. Let Y_a be the twist obtained by gluing $Y|_{U_i}$ using

$$y_j = s(y_i, a_{ij}).$$

Proposition 4.1. *There is a common complex manifold W and holomorphic affine bundles*

$$Y \longleftarrow W \longrightarrow Y_a$$

with fibre $\mathbb{C}^{\text{rank } E}$. Consequently Y is Oka if and only if Y_a is Oka.

Proof. We set $W = P \times_B Y$. The projection $q_0 : W \rightarrow Y$ is the pulled-back affine bundle, so W is a manifold even if π has critical points. Define

$$q_a(p, y) = [s(y, p_i)] \in Y_a.$$

On overlaps the action law gives

$$s(y, p_j) = s(s(y, p_i), a_{ij}),$$

which is precisely the gluing relation of Y_a ; hence q_a is well defined and holomorphic. Over U_i it has the explicit bundle coordinates

$$(p, y) \longmapsto (z_i = s(y, p_i), p_i), \quad (z_i, p_i) \longmapsto (p, y = s(z_i, -p_i)).$$

These are inverse holomorphic maps, the first coordinate changes by the twist transition, and the second by an affine translation, so q_a is an affine bundle; the same formulas prove submersivity at points fixed by the action. Since an affine bundle is a fibre bundle with fibre $\mathbb{C}^{\text{rank } E}$, the Oka equivalence follows twice from the fibre-bundle theorem [6, Theorem 1.4]. $\square$

The descent can also be seen directly from convex approximation, without invoking the fibre-bundle theorem. A holomorphic map from a Stein neighbourhood of a convex compact set into Y_a lifts to the pulled-back affine bundle by Cartan's theorem B, and an entire approximation into an Oka W , followed by q_a , gives the required approximation into Y_a ; the projection to B is allowed to vary.

4.1 Analytic genus-one K3 torsors

We now apply the proposition to the situation of interest. Let $\pi : J \rightarrow \mathbb{P}^1$ be a Jacobian elliptic K3 surface with zero section e , and let σ be a nonzero holomorphic two-form. We put $E = T^*\mathbb{P}^1$. Symplectic contraction defines

$$A : \pi^*E \longrightarrow T_J, \quad A = \sigma^\sharp \circ d\pi^*.$$

Lemma 3.1 provides an additive action on every complete fibre. To apply the affine-bundle equivalence, we must identify its normalization with the additive sheaf used in the analytic Tate–Shafarevich construction, namely

$$L = R^1\pi_*\mathcal{O}_J \simeq \mathcal{O}_{\mathbb{P}^1}(-2).$$

The normalization has to be checked on the whole base, and here the presence of singular fibres is a genuine issue. The section satisfies $d\pi \circ de = \text{id}$, so it lies in the smooth locus of π even above singular fibres; restricting A to e and taking the normal component therefore gives an everywhere invertible map

$$E \xrightarrow{\sim} N_{e/J}.$$

Now the boundary map for

$$0 \longrightarrow \mathcal{O}_J \longrightarrow \mathcal{O}_J(e) \longrightarrow N_{e/J} \longrightarrow 0$$

identifies $N_{e/J}$ with L over the entire base; see [11, Chapter 11, Section 2.2]. Denote the composite by $\theta : E \xrightarrow{\sim} L$, with the sign fixed by the Abel–Jacobi convention. On a smooth fibre this boundary map is the differential of $P \mapsto \mathcal{O}(P - e)$, and since the action is translation invariant there, uniqueness of the torus exponential shows that translation by v is translation by $\exp(\theta(v))$. In particular, no factor with zeros or poles at a singular value has been inserted.

Theorem 4.2. *All analytic genus-one K3 torsors with a fixed Jacobian J have the same Oka status as J .*

Proof. The analytic Tate–Shafarevich construction gives a surjection

$$H^1(\mathbb{P}^1, L) \simeq H^2(J, \mathcal{O}_J) \longrightarrow \text{Sha}_{\text{an}}(J).$$

Moreover, the image of an additive cocycle is represented by gluing the complete local elliptic surfaces by its exponential translations. One may choose the cover so that each singular value belongs to only one member, in which case all gluing overlaps lie over the smooth-fibre locus. The surjectivity and this gluing construction are given in [11, Chapter 11, Corollary 5.12 and Remark 5.15].

Given a torsor X , choose an additive lift η and a cocycle representing it, and transfer that cocycle by θ^{-1} to E . The preceding normalization shows that its action twist is precisely X on the overlaps, while each singular fibre is unchanged inside its whole local chart. Lemma 3.1 and Proposition 4.1 therefore give a common affine-line bundle for J and X , which proves the equivalence. $\square$

Two remarks on the scope of the argument. The cohomological surjectivity covers the full analytic Tate–Shafarevich group, including non-torsion classes, and depends on $H^3(J, \mathbb{Z}) = 0$; exponential translations are compared on smooth-fibre overlaps, while each complete local elliptic surface is retained, and the Hamiltonian action is defined on its singular fibres as well. Note also that the common total space is the pullback of an affine-line bundle over the base, so that the construction avoids the generally singular fibre product of the two compact elliptic surfaces.

5 The discriminant-three Oka K3 surface

5.1 The surface and the eleven fibrations

Everything in this section is explicit, and this is deliberate: the model below is the one place where the dominating family of sprays can be verified by hand. On $B = \mathbb{P}_X^1 \times \mathbb{P}_Y^1$, with homogeneous coordinates $[X_0 : X_1]$ and $[Y_0 : Y_1]$, let $L = \mathcal{O}_B(1, 1)$ and

$$s = X_0 X_1 (X_1 - X_0) Y_0 Y_1 (Y_1 - Y_0) \in H^0(B, L^3).$$

Let $\pi : W \rightarrow B$ be the cyclic triple cover $Z^3 = s$, where Z is the fibre coordinate of L . Let $X \rightarrow W$ be its minimal resolution. On the affine chart $X_0 Y_0 \neq 0$, write

$$x = X_1/X_0, \quad y = Y_1/Y_0, \quad z = Z/(X_0 Y_0), \quad z^3 = x(x-1)y(y-1). \quad (17)$$

Here X denotes the resolved surface, whereas X_0, X_1 are the homogeneous coordinates on the first factor of B .

For the index set $I = \{0, 1, \infty\}$, the denominator sections are

$$\ell_0(X) = X_1, \quad \ell_1(X) = X_1 - X_0, \quad \ell_\infty(X) = X_0,$$

with the same convention for the second factor. We consider the two projections and the nine rational functions

$$f_x = x, \quad f_y = y, \quad f_{rs} = \frac{Z}{\ell_r(X)\ell_s(Y)}, \quad r, s \in I. \quad (18)$$

Thus the nine affine ratios are $z, z/x, z/(x-1), z/y, z/(y-1)$ and the four ratios with one of $x, x-1$ and one of $y, y-1$ in the denominator.

Domination will be achieved by the subfamily of eight functions

$$\mathcal{F}_8 = \{f_x, f_y\} \cup \{f_{rs} : r, s \in I, r \neq s\}. \quad (19)$$

Theorem 5.1. *The surface X is a projective K3 surface with $\rho(X) = 20$ and $\det \text{NS}(X) = -3$, and is not Kummer. All eleven functions in (18) extend to morphisms $X \rightarrow \mathbb{P}^1$ with connected genus-one general fibres. Their combined map*

$$\Phi : X \longrightarrow (\mathbb{P}^1)^{11}$$

is an immersion. The complete holomorphic line-bundle sprays associated with the eight fibrations in $\mathcal{F}_8$ already form a dominating family. Consequently X is subelliptic and Oka.

Two ingredients go into the proof. The analytic mechanism is the compact-fibre spray construction of Xie–Zhao [21, Proposition 3.2], combined with Forstnerič’s subelliptic Oka theorem [4, Theorem 1.1]; what remains, and what occupies the rest of this section, is an explicit global computation of the fibrations and of their differentials. We should also note that dominability of X alone is not new: since X is elliptic, Buzzard–Lu already give that [2]. What matters here is the stronger spanning property.

5.2 The compact K3 model and its lattice

The branch divisor of π consists of the three vertical and three horizontal lines indexed by I . Their nine crossings are the only singularities of W , each analytically of the form $ab = w^3$ and hence of type A_2 ; the cover is connected and normal, and it is projective because it is finite over B .

The cyclic-cover canonical formula gives

$$K_W = \pi^*(K_B + 2L) = \mathcal{O}_W.$$

The A_2 resolutions are crepant, and since

$$\pi_* \mathcal{O}_W = \mathcal{O}_B \oplus \mathcal{O}_B(-1, -1) \oplus \mathcal{O}_B(-2, -2),$$

Künneth gives $H^1(W, \mathcal{O}_W) = 0$. Rational double points preserve this group under resolution, so $H^1(X, \mathcal{O}_X) = 0$ and $K_X = \mathcal{O}_X$; thus X is a K3 surface.

The first projection has smooth genus-one fibres away from $x = 0, 1, \infty$. A general fibre is a connected triple cover of $\mathbb{P}_y^1$, fully branched at three points, while each of the three special fibres has a central multiplicity-three rational curve and three arms of multiplicities two then one; this follows from the resolution charts in (21), and it identifies the fibre as Kodaira type IV*. The strict transform of $y = \infty$ is a section.

The section, the fibre and the fibre components therefore give a sublattice

$$U \oplus E_6(-1)^{\oplus 3} \subset \text{NS}(X)$$

of rank twenty and determinant -27 ; since a complex K3 has Picard number at most twenty, $\rho(X) = 20$, and the Mordell–Weil group has rank zero by the Shioda–Tate formula [16, Corollary 6.13].

We next determine the discriminant. On the generic plane cubic, the three points with $z = 0$, at $y = 0, 1, \infty$, are distinct flexes, so taking the last as zero and either of the other two gives a nonzero three-torsion section. Let t be the order of the finite Mordell–Weil group. The elliptic-surface exact sequence and index formula [16, Theorem 6.3 and (19)] give

$$|\det \text{NS}(X)| = 27/t^2.$$

Here $3 \mid t$, while integrality forces $t^2 \mid 27$; hence $t = 3$ and $\det \text{NS}(X) = -3$.

By unimodularity of the K3 lattice, the transcendental lattice has rank two and determinant three. This is the discriminant-three singular K3, which also appears in the equal- $j = 0$ Inose case in [16, Table 5]. Note finally that X is not a Kummer surface: the Kummer relation $T(\text{Kum}(A)) = T(A)(2)$ [16, §13.3] would make a rank-two transcendental determinant divisible by four.

5.3 Local coefficients and extension of the fibrations

Coordinates at the branch divisor

Near a branch index $i \in I$, trivialize $\mathcal{O}_{\mathbb{P}^1}(1)$ by $e_i = X_0$ for $i = 0, 1$, and by $e_\infty = X_1$. We use the local coordinate

$$a = x \ (i = 0), \quad a = x - 1 \ (i = 1), \quad a = 1/x \ (i = \infty).$$

There are functions q_i and units D_{ir} such that

$$\frac{X_0 X_1 (X_1 - X_0)}{e_i^3} = a q_i(a), \quad \frac{\ell_r(X)}{e_i} = a^{[r=i]} D_{ir}(a).$$

Here $[r = i]$ is one when $r = i$ and zero otherwise. The coefficients are

$$\begin{array}{c|ccc} i & q_i(a) & D_{i0}(a) & D_{i1}(a) & D_{i\infty}(a) \\ \hline 0 & a - 1 & 1 & a - 1 & 1 \\ 1 & a + 1 & a + 1 & 1 & 1 \\ \infty & 1 - a & 1 & 1 - a & 1 \end{array} \quad . \tag{20}$$

We use the same convention, with coordinate b , at a branch index j in the second factor. If $\zeta = Z/(e_i e_j)$, then

$$\zeta^3 = abq_i(a)q_j(b).$$

We choose a local cube root of the unit $q_i(a)q_j(b)$ and put $w = \zeta/(q_i(a)q_j(b))^{1/3}$; the local equation is then $ab = w^3$.

For clarity, the infinity transitions in affine notation are

$$\begin{aligned} a = 1/x, \quad \zeta = z/x : & \quad \zeta^3 = a(1-a)y(y-1), \\ b = 1/y, \quad \eta = z/y : & \quad \eta^3 = x(x-1)b(1-b), \\ a = 1/x, \quad b = 1/y, \quad \theta = z/(xy) : & \quad \theta^3 = a(1-a)b(1-b). \end{aligned}$$

These transitions describe the same projective cyclic cover along the boundary.

The resolution and the ratio table

The minimal resolution of $ab = w^3$ has charts

	a	b	w	
C_0	u	$u^2 v^3$	uv	
C_1	$u^2 v$	uv^2	uv	
C_2	$u^3 v^2$	v	uv	(21)

The two exceptional curves meet at $(0,0)$ in C_1 , while the charts C_0, C_2 include their outer intersections with the strict transforms of the branch curves. These are the charts of the ordinary blowup of $ab = w^3$ at the origin: in the a -chart, $b = a^2 v^3$ and $w = av$; in the w -chart, writing $a = wu$ and $b = wv$ gives $w = uv$; and the b -chart is symmetric. Thus the displayed modification is proper and smooth and resolves the singularity. In each chart, the relation $ab = w^3$ holds, and direct differentiation gives

$$\frac{da \wedge db}{w^2} = 3 du \wedge dv.$$

In particular the form $dx \wedge dy/z^2$ extends as a nowhere-zero form through every exceptional chart, up to the nonzero units in (20).

At the crossing (i, j) , let $m = [r = i]$ and $n = [s = j]$. The exact expression for a ratio is

$$f_{rs} = H_{rs}(a, b) \frac{w}{a^m b^n}, \quad H_{rs}(a, b) = \frac{(q_i(a)q_j(b))^{1/3}}{D_{ir}(a)D_{js}(b)}. \quad (22)$$

The function H_{rs} is a holomorphic unit. It is pulled back from the base, so its restriction to each exceptional curve is the constant $H_{rs}(0, 0)$.

The four possible monomial factors in (22) are

	$(m, n) = (0, 0)$	$(1, 0)$	$(0, 1)$	$(1, 1)$	
C_0	uv	v	$(uv^2)^{-1}$	$(u^2 v^2)^{-1}$	
C_1	uv	u^{-1}	v^{-1}	$(u^2 v^2)^{-1}$	
C_2	uv	$(u^2 v)^{-1}$	u	$(u^2 v^2)^{-1}$	(23)

After multiplication by the unit H_{rs} , each entry is holomorphic or is the reciprocal of a holomorphic function; either way it defines a morphism to $\mathbb{P}^1$ throughout the chart. This settles extension on every exceptional component and at all their intersection points.

It remains to look at smooth points of the branch curves. At such a point a transverse uniformizer w has base coordinate a unit times w^3 , so a ratio is a unit times w if its

denominator does not vanish, and a unit times w^{-2} if it does; away from the branch divisor there is no denominator zero. This proves the global extension of every f_{rs} on the minimal surface X , without any additional source blowup, while the projections are already morphisms by construction.

5.4 The genus-one fibres and their symmetries

The projection fibrations have genus-one general fibres by the three-branch-point calculation above. For $z = c$, the fibre is the double cover of $\mathbb{P}_x^1$ with discriminant

$$\frac{x^2 - x + 4c^3}{x(x-1)}.$$

For generic nonzero c the two numerator roots, zero and one are distinct, so there are four simple branch points and the fibre is connected of genus one.

The remaining eight ratios have the same generic fibre, and the reason is a family of explicit automorphisms. We use the following Möbius transformations and accompanying factors:

r	$\phi_r(x)$	$Q_r(x)$
∞	x	1
0	$1/x$	$-1/x$
1	$1/(1-x)$	$1/(x-1)$

(24)

Then

$$\phi_r(x)(\phi_r(x) - 1) = Q_r(x)^3 x(x-1).$$

Consequently

$$(x, y, z) \longmapsto (\phi_r(x), \phi_s(y), Q_r(x)Q_s(y)z) \tag{25}$$

is an automorphism of W . Indeed, each Möbius transformation permutes the three branch points and takes their homogeneous binary cubic to a nonzero scalar multiple, while the factor Q_r is the corresponding line-bundle transition with a chosen cube-root scalar; for example, under $x \mapsto 1/x$, $z \mapsto -z/x$, the target infinity-chart coordinate is $(-z/x)/(1/x) = -z$, which is holomorphic at $x = 0$, and the same check applies on the other charts.

These automorphisms lift to the minimal resolution of the rational double points, and the pullback of z under (25) is a nonzero constant times f_{rs} . Hence all nine ratios have connected genus-one general fibres, and together with the extension calculation above this establishes the required fibration properties of all eleven functions.

5.5 The differential spanning calculation

It remains to check that the differentials of the eight functions in $\mathcal{F}_8$ span at every point of the compact surface, which will also prove that the full product map Φ is an immersion. The verification is a case check, and the bookkeeping that makes it short is the following. The six retained ratio indices give two entries in every row and every column of $I \times I$, so for every branch crossing (i, j) one of the two entries in row i has $s \neq j$, and one of the two entries in column j has $r \neq i$. Thus the retained family provides a ratio with $r = i$, $s \neq j$ and another with $r \neq i$, $s = j$ at all nine crossings, and these are the choices used in the exceptional charts below.

Away from the branch divisor

Here the cyclic cover is unramified, so the two projection differentials span the cotangent space.

Smooth points of the branch curves

We consider a point over a vertical branch line $x = i$, away from the horizontal branch lines. There are local coordinates (w, b) on X , with the second projection given by b . We take $r \neq i$ and then $s \neq r$, so that f_{rs} belongs to $\mathcal{F}_8$; since the point lies away from all horizontal branch lines, the second denominator is nonzero there, and the ratio is $f_{rs} = wH$ with H a holomorphic unit. At $w = 0$, its differential has nonzero dw coefficient and is independent of db . Interchanging the factors preserves the selected family and gives the horizontal case, and the projective coordinates above cover the branch lines at infinity as well.

Outer exceptional charts

In C_0 , the first projection has local target coordinate $a = u$. We choose a retained pair with $r = i$ and $s \neq j$ by the row argument above, so (23) gives $f_{rs} = Hv$. Along the exceptional curve $u = 0$, the coefficient H restricts to the nonzero constant $H_0 = H(0, 0)$, so

$$da \wedge df_{rs} = H_0 du \wedge dv \neq 0.$$

This includes the outer endpoint $v = 0$. In C_2 , the second projection has local coordinate $b = v$; the retained column choice with $r \neq i$ and $s = j$ gives $f_{rs} = Hu$, and

$$db \wedge df_{rs} = -H_0 du \wedge dv \neq 0$$

along the exceptional curve $v = 0$, including its outer endpoint.

The middle exceptional chart

We choose retained ratios with $r = i, s \neq j$ and $r' \neq i, s' = j$ by the same row and column argument. They have expressions H/u and K/v , so their reciprocal target coordinates near the exceptional locus are

$$\alpha = u/H, \quad \beta = v/K,$$

where H, K are units pulled back from $a = u^2v, b = uv^2$. At the middle intersection $(0, 0)$,

$$d\alpha = H_0^{-1}du, \quad d\beta = K_0^{-1}dv.$$

They are independent. Along $u = 0$, the first differential is $H_0^{-1}du$, while the second has dv coefficient K_0^{-1} , since K is constant in the tangent v -direction there; its possible additional du term does not affect the wedge. Interchanging u, v handles the other exceptional axis, so

$$d\alpha \wedge d\beta = (H_0K_0)^{-1}du \wedge dv \neq 0$$

along both axes. Here we have kept the actual coefficient units rather than replacing them by constants off the exceptional curves.

The open unramified locus, the smooth branch points and these three charts cover X , so the differentials of the selected eight functions span at every point, including all higher-multiplicity fibre components and all their intersections; consequently the full eleven-function map Φ is an immersion.

5.6 The Oka property

We fix a nonzero holomorphic two-form σ on the K3 surface X . By Lemma 3.1, each fibration has an entire spray with vertical differential $A_f = \sigma^\sharp \circ df^*$, and the preceding differential calculation gives

$$\sum_{f \in \mathcal{F}_8} \text{im } A_{f,x} = T_x X \quad (x \in X).$$

These eight line-bundle sprays therefore form a finite dominating family, so X is subelliptic and is Oka by [4, Theorem 1.1 and p. 529], which proves Theorem 5.1.

6 Compact Inose models and Picard rank twenty

The seed surface of the previous section is not yet enough, because a general genus-one K3 surface need not be related to it by any obvious deformation; what we need is a family of compact Oka models that is rich enough to match the exceptional geometry of an arbitrary surface, and the Inose fibrations provide exactly that.

Theorem 6.1. *The smooth minimal compact K3 model*

$$X_{A,B} : \quad y^2 = x^3 - 3At^4x + t^5(t^2 - 2Bt + 1) \quad (26)$$

is Oka for every $(A, B) \in \mathbb{C}^2$. Every complex K3 surface of Picard rank twenty is Oka.

K3 surfaces of Picard rank twenty are traditionally called *singular K3 surfaces*, and we emphasise that all compact surfaces in the theorem are smooth. The parameter range includes the equal-factor and isogenous complex multiplication cases, as well as discriminants three and four; as for the Picard-rank-twenty assertion, it rests on the Shioda–Inose construction and classification, which provide an isomorphism of compact surfaces rather than an isogeny.

The reverse Kummer companion is

$$S_{A,B} : \quad v^2 = u^3 - 3As^4u + s^4(s^4 - 2Bs^2 + 1), \quad \nu_{A,B}(s, u, v) = (-s, u, -v), \quad (27)$$

where the smooth minimal compact model is understood. The proof compares these companions with the seed at $(A, B) = (0, 1)$, and it needs two inputs. We first prove that the complement of the fixed points is Oka on every product Kummer companion; we then identify its marked involution and construct an equivariant diffeomorphism to the seed which is biholomorphic near the fixed points. These are precisely the two inputs required by Corollary 2.7.

6.1 Every product reverse-Nikulin complement is Oka

Proposition 6.2. *Let $S = \text{Kum}(E_1 \times E_2)$ for two complex elliptic curves, with a zero section of the fibration $s = y_2/y_1$ arising from a finite two-torsion pair. Denote by ν the base-deck involution $s \mapsto -s$ composed with fibrewise inversion, and by P its fixed set. Then $S \setminus P$ is Oka.*

We choose Legendre coordinates

$$\begin{aligned} E_1 : \quad y_1^2 &= x_1(x_1 - 1)(x_1 - \mu), \\ E_2 : \quad y_2^2 &= x_2(x_2 - 1)(x_2 - \lambda), \quad \mu, \lambda \notin \{0, 1, -1\}, \end{aligned} \quad (28)$$

and take $O = E_{0,0}$; every elliptic curve admits such coordinates. Two normalizations will be used below without further comment. The affine substitution $x' = 1 - x$, $y' = iy$ changes a Legendre parameter -1 to 2 : it preserves the pole at infinity and changes s by a constant factor. The assertion is also unchanged when the chosen finite two-torsion origin is replaced, since for the difference section Q , which is rational over s^2 , the deck fixes Q and

$$t_Q \nu t_{-Q} = t_{2Q} \nu$$

is deck composed with inversion about Q , while translation extends to an automorphism of the compact minimal elliptic K3 model.

We write

$$I = \{\infty, 0, 1, \mu\}, \quad J = \{\infty, 0, 1, \lambda\}, \quad d_\mu = \frac{2\mu}{1+\mu}, \quad d_\lambda = \frac{2\lambda}{1+\lambda}.$$

The values d_μ, d_λ are finite and nonzero, and each is distinct from the finite branch values of its factor. Denote by F_p the image of $\{p\} \times E_2$, with coordinate x_2 , and by G_q the image of $E_1 \times \{q\}$, with coordinate x_1 . The standard exceptional curves are $E_{p,q}$, with endpoints

$$A_{p,q} = F_p \cap E_{p,q}, \quad C_{p,q} = G_q \cap E_{p,q}.$$

The twenty-four basic curves form the double Kummer configuration with edges $F_p - E_{p,q} - G_q$.

Lemma 6.3. *The fixed set of the specified involution is*

$$P = \{A_{\infty,0}, C_{\infty,0}, C_{0,0}, C_{0,\infty}, A_{0,\infty}, A_{0,0}, \\ (x_2 = d_\lambda) \text{ on } F_\infty, (x_1 = d_\mu) \text{ on } G_\infty\}. \quad (29)$$

The two projection images are $\{\infty, 0, d_\mu\}$ and $\{\infty, 0, d_\lambda\}$.

Proof. The plane cubic

$$x_2(x_2 - 1)(x_2 - \lambda) = s^2 x_1(x_1 - 1)(x_1 - \mu) \quad (30)$$

has origin $(0,0)$ already over s^2 , so we may choose Jacobian coordinates in which the base deck fixes the fibre coordinates. Both the deck and inversion negate the holomorphic two-form and commute, hence ν is symplectic, and its fixed points lie over $s = 0, \infty$, where the fibres have type IV* [13, Example 1.2].

The component group of a IV* fibre is $\mathbb{Z}/3\mathbb{Z}$ [16, Lemma 7.3], and inversion preserves the identity leg while exchanging the other two. On the central components its actions are determined by their three attachment values:

$$F_\infty : x_2 \mapsto \frac{\lambda x_2}{(1+\lambda)x_2 - \lambda}, \quad G_\infty : x_1 \mapsto \frac{\mu x_1}{(1+\mu)x_1 - \mu}. \quad (31)$$

The first map fixes 0, exchanges 1 and λ , and has d_λ as its second fixed point. The identity outer components have actions

$$G_0 : x_1 \mapsto -x_1, \quad F_0 : x_2 \mapsto -x_2. \quad (32)$$

Each is a nontrivial involution fixing the origin at coordinate zero and the node at infinity, and neither can act identically on its component: a nonidentity symplectic involution cannot fix a curve, since finite-order linearization at a point of that curve would have both eigenvalues one.

On each fibre the fixed points are therefore the two nodes of the identity leg, the zero-section point on its outer component, and the second central fixed point. Every invariant rational component has exactly those two fixed points, while the exchanged legs contribute none, and this gives (29). $\square$

Proof of Proposition 6.2. We construct protected sprays and show that every point of $S \setminus P$ can be moved into a locus where two of them span the tangent space. The projections handle the complement of the basic curves; three translated fibrations and four ratio fibrations handle the remaining points.

Begin with the spanning locus and the first moves. We protect $\{\infty, 0, d_\mu\}$ for x_1 and $\{\infty, 0, d_\lambda\}$ for x_2 , using (16). Outside the basic configuration the product map (x_1, x_2) is locally biholomorphic, since it is the unramified part of the double cover of $\mathbb{P}^1 \times \mathbb{P}^1$ branched over the four vertical and four horizontal branch lines; away from the protected projection values, the two projection sprays therefore span the tangent space. Denote this locus by U_{good} .

We use the three two-torsion translated s -fibrations with pole-origin pairs (∞, ∞) , $(0, 0)$, and $(1, 1)$, protected at zero and infinity; these pairs specify the origins of the elliptic factors, where their translated y -coordinates have poles, and the zero section of the original cubic remains $O = E_{0,0}$. The first pair gives the original fibration $s = y_2/y_1$. Every F_p and G_q lies in the zero or pole fibre of each translate, while for pole origins $p_0 \neq p$, $q_0 \neq q$ the exceptional curve $E_{p,q}$ is a section, with its endpoints over zero and infinity [13, Example 1.2 and §2.2]. Since the three pairs have pairwise distinct first and second coordinates, a given p excludes at most one pair and a given q excludes at most one more, so at least one of the three fibrations makes $E_{p,q}$ a section. These assertions also follow in the torus blowup charts: at a finite pair the ratio y_2/y_1 has degree one on the exceptional line; at (∞, q) it is a unit times $z_1^3 z_2$, giving multiplicity two after passage to the quotient; and the central and outer multiplicities are three and one. Translation gives every choice of origins.

At an interior point of an exceptional curve, choose a translate for which that curve is a section. Its differential is nonzero on the curve, so its Hamiltonian direction is transverse to it, and in a resolved node chart both branch-coordinate differences are nonzero multiples of a normal coordinate at an interior point; a small nonzero flow therefore moves both projections away from their branch values and into U_{good} .

Outside the basic curves a point on just one of $x_1 = d_\mu$, $x_2 = d_\lambda$ escapes by the other projection spray. At either point over (d_μ, d_λ) , both partial derivatives of s are nonzero: indeed, for $f_z(x) = x(x-1)(x-z)$,

$$f'_z(d_z) = -\frac{3z(z-1)^2}{(1+z)^2} \neq 0, \quad (33)$$

and differentiating (30) gives the assertion. Here s is finite and nonzero, so its protected spray is available, and its Hamiltonian direction changes both projection coordinates and reaches U_{good} . This includes all special equal-factor or isogenous parameters.

We turn to the residual points. On F_p away from its nodes, the x_2 -spray is transverse and available except at $x_2 = d_\lambda$, while the other protected values are nodes; the symmetric statement holds on G_q . At $A_{p,q}$, dx_2 is nonzero and its Hamiltonian direction is tangent to $E_{p,q}$, so it is available for $q \in \{1, \lambda\}$, and at $C_{p,q}$ the corresponding dx_1 direction is available for $p \in \{1, \mu\}$; the standard resolved node chart verifies these endpoint statements.

After deleting (29), the complete residual list is

$$\begin{aligned} & A_{1,0}, A_{\mu,0}, A_{1,\infty}, A_{\mu,\infty}, A_{\infty,\infty}, \\ & C_{\infty,1}, C_{\infty,\lambda}, C_{0,1}, C_{0,\lambda}, C_{\infty,\infty}, \\ & (x_2 = d_\lambda) \text{ on } F_0, F_1, F_\mu, \quad (x_1 = d_\mu) \text{ on } G_0, G_1, G_\lambda. \end{aligned} \tag{34}$$

There are ten nodes and six nonnodal points. Additional reducible fibres introduce no further exception, because the preceding argument uses the product map and applies to every point outside the basic configuration.

Four ratio fibrations now handle twelve points of the residual list:

$$\begin{aligned} R_+ &= x_1 \frac{x_2 - \lambda}{x_2 - 1}, & R_- &= x_1 \frac{x_2 - 1}{x_2 - \lambda}, \\ T_+ &= x_2 \frac{x_1 - \mu}{x_1 - 1}, & T_- &= x_2 \frac{x_1 - 1}{x_1 - \mu}. \end{aligned} \tag{35}$$

These are genuine J_6 fibrations [13, §2.4]. Directly, a ratio of two Möbius branch coordinates extends across the resolved branch crossings where $0/0$ or ∞/∞ occurs; and a generic $(1, 1)$ level curve of the ratio passes through two branch crossings, leaving four distinct simple branch points on its double cover, so its genus is one. Note that this reasoning does not require nonisogenous factors or generic parameters.

To ensure that these sprays preserve $S \setminus P$, evaluate the fibrations on the eight fixed points in (29). Their complete sets of values are

$$R_+(P) = R_-(P) = \{0, \infty, d_\mu\}, \quad T_+(P) = T_-(P) = \{0, \infty, d_\lambda\}. \tag{36}$$

For $R_\pm$, the endpoints on $E_{\infty,0}$ and the nonnodal point on F_∞ have value infinity; the endpoints on $E_{0,0}$ and $E_{0,\infty}$ have value zero; and the point on G_∞ has value d_μ , with no indeterminate ratio occurring at a puncture. The same computation applied to $T_\pm$ in place of $R_\pm$ gives the second assertion, and we protect these complete fibres.

For $p \in \{1, \mu\}$, with p' the other value,

$$d_\mu = \frac{2pp'}{p + p'}, \quad p \neq d_\mu.$$

At $A_{p,0}$ the two R -values are $p\lambda$ and p/λ ; if both were protected, each would equal d_μ , forcing $\lambda^2 = 1$, which is excluded. At $A_{p,\infty}$ both values are p and hence unprotected, while at the point $x_2 = d_\lambda$ on F_p the two values are $-p\lambda$ and $-p/\lambda$, and simultaneous protection again forces $\lambda^2 = 1$. Thus at least one of the two sprays is available at each of these six points.

The differential is nonzero at every indicated endpoint. If a ratio $\varphi(x_1)/\psi(x_2)$ has finite nonzero factors at (p, q) , a resolved chart at $A_{p,q}$ has branch coordinates $\alpha = \rho\theta^2$, $\beta = \rho$, with exceptional curve $\rho = 0$ and endpoint $\theta = 0$, and its expansion is

$$\frac{\varphi(p)}{\psi(q)} + \rho(c\theta^2 - d) + O(\rho^2), \quad d \neq 0.$$

The Hamiltonian direction points into the exceptional interior; interchanging the two coordinates gives the other endpoint, the case of infinite branch coordinates included. At the nonnodal point on F_p , the ratio restricts to a degree-one coordinate, so its direction is transverse to F_p . The moves already constructed then carry the point into U_{good} .

For $q \in \{1, \lambda\}$ the symmetric T -values are $q\mu, q/\mu$ at $C_{0,q}$, both q at $C_{\infty,q}$, and $-q\mu, -q/\mu$ at the point $x_1 = d_\mu$ on G_q . The only possible simultaneous collision is $\mu^2 = 1$,

which is excluded, and $q \neq d_\lambda$; the same differential calculation handles these six points. Thus every parameter collision among all twelve points has been checked.

It remains to deal with the last four points. We apply the actual involution to the four points still remaining in (34), using (31)–(32): their image coordinates are respectively $d_\lambda/2$ on F_∞ , $d_\mu/2$ on G_∞ , $-d_\lambda$ on F_0 , and $-d_\mu$ on G_0 . All are finite, nonzero, and different from the corresponding protected d -value, so the projection sprays handle the images.

The nine entire protected sprays—two projections, three translated s -fibrations, and four ratio fibrations—and ν therefore move every point of $S \setminus P$ into U_{good} , and the orbit criterion in Section 3 proves that $S \setminus P$ is Oka. $\square$

6.2 The exact marked Inose companions

We now identify the product companion with (27), retaining the involution and the zero section at every parameter. The computation uses the explicit product equation of Kuwata–Shioda [13, §2.2].

For $z \notin \{0, 1\}$ write

$$Q_z = z^2 - z + 1, \quad D_z = z^2(z - 1)^2, \quad L_z = (z - 2)(z + 1)(2z - 1), \quad J_z = \frac{4Q_z^3}{27D_z}.$$

Here $J_z = j(E_z)/1728$. With origin $(0, 0)$, the Weierstrass equation of (30) is

$$\begin{aligned} Y^2 &= X^3 + as^2X^2 + bs^4X + 16D_\mu s^8 + cs^6 + 16D_\lambda s^4, \\ a &= 4(\mu + 1)(\lambda + 1), \\ b &= 16((\mu\lambda + 1)(\mu + \lambda + 1) - 1), \\ c &= 32\mu\lambda(\mu + 1)(\lambda + 1). \end{aligned}$$

The displayed coordinate formulas in [13] use only even powers of s , so the deck fixes X, Y , and they define a degree-one identification with the Weierstrass generic fibre at every allowed parameter. Let us justify that identification, since it is what forces the marked involution to be the one we want. Near the origin, $x_2 = (\mu s^2/\lambda)x_1 + O(x_1^2)$, and those formulas give leading terms $4\lambda^2/x_1^2$ for X and $-8\lambda^3/x_1^3$ for Y ; moreover X has no other pole on the generic cubic. At the other zeros of x_1x_2 , its numerator vanishes and cancels the simple denominator zero, while at infinity the numerator and denominator have the same degree and both leading coordinates are nonzero. Thus its polar divisor is exactly $2O$, and the resulting map to the smooth Weierstrass generic fibre has degree one, since the pullback of that fibre's degree-two x -coordinate has degree two. This also covers equal or isogenous elliptic factors.

The substitution $X' = X + (a/3)s^2$ gives

$$Y^2 = X'^3 - \frac{16}{3}Q_\mu Q_\lambda s^4 X' + 16D_\mu s^8 + \frac{32}{27}L_\mu L_\lambda s^6 + 16D_\lambda s^4. \quad (37)$$

We choose nonzero constants α, γ satisfying

$$\alpha^4 = D_\mu/D_\lambda, \quad \gamma^6 = 16D_\lambda/\alpha^4.$$

Then $\sigma = \alpha s$, $u = X'/\gamma^2$, $v = Y/\gamma^3$ gives (27), with

$$A = \frac{16Q_\mu Q_\lambda}{9\alpha^4\gamma^4}, \quad B = -\frac{L_\mu L_\lambda}{27D_\lambda\alpha^2}. \quad (38)$$

Since $L_z^2 = 4Q_z^3 - 27D_z$,

$$A^3 = J_\mu J_\lambda, \quad B^2 = (1 - J_\mu)(1 - J_\lambda). \quad (39)$$

Conversely, for any prescribed A, B , choose elliptic factors with normalized invariants equal to the roots of

$$J^2 - (A^3 - B^2 + 1)J + A^3 = 0.$$

We choose their Legendre parameters outside $\{0, 1, -1\}$. The constants in (38) then have the prescribed cube and square: multiplication of α by i changes the sign of B without changing A , while multiplication of γ by sixth roots of unity provides all cube-root choices of A without changing B . Thus every finite pair (A, B) is obtained.

The final short Weierstrass coordinates preserve the zero section and the deck action. Fibrewise inversion is $(u, v) \mapsto (u, -v)$, so the marked reverse involution is exactly $(-\sigma, u, -v)$, and uniqueness of the relatively minimal elliptic model extends the generic identification over all singular components. Proposition 6.2 therefore proves the Oka property of the actual punctured companion for every (26).

6.3 A smooth equivariant family of the marked pairs

Proposition 6.4. *For every $(A, B) \in \mathbb{C}^2$, the pair $(S_{A,B}, \nu_{A,B})$ is equivariantly diffeomorphic to $(S_{0,1}, \nu_{0,1})$ by a diffeomorphism biholomorphic near all eight fixed points.*

Proof. The idea is to resolve the companion family over a curve that joins the seed to the target, choosing the resolutions in pairs under the involution, and then to transport between the endpoint fibres along relative coordinates that are kept constant at the fixed sections.

Let us first describe the singularities of the family. The Weierstrass models (27), with line bundle $\mathcal{O}_{\mathbb{P}^1}(2)$, form a proper family, and at the zero section the projective Weierstrass chart is smooth. Near $s = 0$, the function

$$h(s, u, A, B) = 1 - 3Au - 2Bs^2 + s^4$$

is a unit equal to one at the singular section, so a holomorphic fourth root gives the coordinate

$$r = sh(s, u, A, B)^{1/4}.$$

Its derivative in s at the singular section is one, so it is a legitimate relative coordinate even though it depends on u . The equation becomes $v^2 = u^3 + r^4$, with involution $(r, u, v) \mapsto (-r, u, -v)$; this is an equivariantly constant E_6 singularity. The infinity chart $s' = 1/s$, $u' = u/s^4$, $v' = v/s^6$ gives the same equation, and resolving these two constant germs by their minimal resolutions times the parameter domain resolves them, the constant automorphism lifting uniquely.

On $s \neq 0$, coordinates $\xi = u/s^2$, $\eta = v/s^3$ give

$$\eta^2 = \xi^3 - 3A\xi + s^2 + s^{-2} - 2B, \quad \nu(s, \xi, \eta) = (-s, \xi, \eta). \quad (40)$$

The surface singularities satisfy

$$\eta = 0, \quad \xi^2 = A, \quad s^4 = 1, \quad B = s^2 - \xi^3. \quad (41)$$

For $\epsilon = s^2 \in \{1, -1\}$, the parameter discriminant is

$$(B - \epsilon)^2 = A^3. \quad (42)$$

When $A \neq 0$ the Hessian in ξ, η, s is nondegenerate, so the singularity is A_1 , whereas for $A = 0$ it is A_2 , at $B = \epsilon$. An intersection of the two discriminant curves, namely $B = 0$, $A^3 = 1$, gives two separate exchanged pairs of A_1 points, and there are no further surface singularities.

Near $s^2 = \epsilon$, the coordinate $z = s - \epsilon/s$ has derivative two at the indicated point and satisfies $z^2 = s^2 + s^{-2} - 2\epsilon$. With $U = \eta - z$, $V = \eta + z$, equation (40) becomes

$$UV = \xi^3 - 3A\xi + 2(\epsilon - B). \quad (43)$$

The involution exchanges the two s -neighbourhoods and sends (U, V, ξ) to (V, U, ξ) .

We next resolve these singularities equivariantly and simultaneously. Suppose that, over a smooth curve base, the cubic in (43) has holomorphic ordered roots r_1, r_2, r_3 . Blow up $(U, \xi - r_1)$; its U -chart is

$$\xi = r_1 + Ua, \quad V = a(\xi - r_2)(\xi - r_3),$$

with smooth coordinates (U, a, τ) , where τ is a local base coordinate, and the other chart is

$$U = (\xi - r_1)b_1, \quad b_1V = (\xi - r_2)(\xi - r_3).$$

Blow up $(b_1, \xi - r_2)$; its two charts are

$$\begin{aligned} \xi &= r_2 + b_1c, & V &= c(\xi - r_3), \\ b_1 &= (\xi - r_2)d, & \xi &= r_3 + dV, \end{aligned}$$

with smooth coordinates (b_1, c, τ) and (d, V, τ) , and each projection to the base is a submersion. Distinct roots give Cartier centres and an isomorphism, a double root gives the minimal A_1 resolution with any remaining blowup Cartier, and a triple root gives the minimal A_2 chain; thus the construction inserts exactly the required minimal exceptional configurations.

We choose this resolution in the neighbourhoods of $s = +1, +i$, and its image under $(U, V, \xi) \mapsto (V, U, \xi)$ in the corresponding neighbourhoods of $s = -1, -i$. This produces a holomorphic involution of the resolved total space, the choices of resolution being paired at exchanged singularities.

It remains to build a common curve base over which both cubics split. For a target $(A_*, B_*) \neq (0, 1)$ take

$$A(\tau) = \tau A_*, \quad B(\tau) = 1 + \tau(B_* - 1).$$

Neither polynomial $(B(\tau) - \epsilon)^2 - A(\tau)^3$ is identically zero, so their zeros are finite. Normalize the τ -line in a splitting field of both cubics

$$\xi^3 - 3A(\tau)\xi + 2(1 - B(\tau)), \quad \xi^3 - 3A(\tau)\xi + 2(-1 - B(\tau)). \quad (44)$$

This is a connected smooth algebraic curve, finite and surjective over the line, and all six ordered roots are holomorphic on it: they are integral over $\mathbb{C}[\tau]$ and belong to the integrally closed normalized local rings. Equivalently one can take a connected normalized component of the simultaneous ordered-root cover that dominates the line.

We choose a point over each of $\tau = 0, 1$ and an embedded arc joining them while avoiding all other discriminant preimages; this is possible because removing finitely many points from a connected Riemann surface leaves it path connected. A small simply connected Stein neighbourhood T of the arc avoids those other points and retains global root labels

for both cubics, and it includes target parameters on both discriminant curves as well as the triple-root seed.

We pull back the proper Weierstrass family to T . The two constant E_6 resolutions and the paired local modifications above are proper and are isomorphisms away from the singular loci, their centres there being Cartier, so they glue with the unchanged model on the overlap regions and give a proper holomorphic family $\pi : \mathcal{S} \rightarrow T$ with a holomorphic involution. The displayed charts make it a smooth submersion along every exceptional locus, and it is already a submersion elsewhere. Each fibre is the minimal K3 resolution, so the endpoint fibres and their lifted involutions are precisely the marked pairs under consideration; when the target equals the seed, only the identity map is required.

Finally we transport along the arc with fixed holomorphic coordinates. The form $ds \wedge du/v$ is invariant, so every fibre involution is nontrivial and symplectic, and such an involution has eight isolated fixed points with tangent action $-\text{id}$. One can count them by the holomorphic Lefschetz formula: the trace on $H^0(\mathcal{O}) \oplus H^2(\mathcal{O})$ is two, and each isolated point contributes $1/4$. Its relative fixed locus is finite étale over T and hence consists of eight disjoint holomorphic sections.

Near these sections the normal bundles are trivial, since T is contractible and Stein, and Siu's Stein-neighbourhood theorem together with Cartan extension gives relative holomorphic coordinates with prescribed independent first derivatives. Taking their odd parts under the involution retains these derivatives and makes the action $\xi \mapsto -\xi$, after which the inverse function theorem, uniformly along the compact arc, gives eight disjoint coordinate neighbourhoods.

We prescribe now a horizontal vector field along the arc which keeps these coordinates constant in smaller neighbourhoods of the fixed sections, extend it smoothly by a partition of unity, and average under the involution; the prescribed local part is already invariant. Properness ensures that the flow exists for the full interval, and the endpoint diffeomorphism is equivariant and is the identity in those holomorphic relative coordinates, hence is biholomorphic near every fixed point. $\square$

6.4 The seed quotient and all compact Inose surfaces

We can now recognize the seed of Section 5 among the companions. The invariant functions

$$(t, x, y) = (s^2, s^2u, s^3v) \tag{45}$$

satisfy (26) and, conversely, adjoining s with $s^2 = t$ recovers $u = x/t$ and $v = y/s^3$ generically; this is the exact quotient function field. Blowing up the eight fixed points and taking the quotient resolves its ordinary double points, with exceptional self-intersections -2 , and since birational smooth K3 models are isomorphic, this identifies the complete compact quotient with $X_{A,B}$.

Lemma 6.5. *The quotient at $(A, B) = (0, 1)$ is isomorphic to X .*

Proof. Its equation is

$$y^2 = x^3 + t^5(t - 1)^2.$$

It has fibres II^* at zero and infinity and IV at one: the orders of a_6 are 5, 5, 2, and the discriminant orders are 10, 10, 4. The infinity coordinates $r = 1/t$, $x' = r^4x$, $y' = r^6y$ give $y'^2 = x'^3 + r^5(1 - r)^2$. The line bundle is $\mathcal{O}(2)$, so the elliptic canonical bundle and Leray formulas give the K3 model.

The fibre, section, and nonidentity components give the rank-twenty sublattice

$$U \oplus E_8(-1)^{\oplus 2} \oplus A_2(-1) \subset \text{NS}(X_{0,1})$$

of determinant -3 . An index m would give determinant $-3/m^2$ for the full integral Néron–Severi lattice, forcing $m = 1$; the transcendental lattice is therefore positive even of rank two and determinant three.

There is a unique oriented isometry class of such lattices. A shortest primitive vector extended to a positively oriented reduced basis gives a matrix

$$\begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix}, \quad |b| \leq a \leq c.$$

Then $3 = 4ac - b^2 \geq 3a^2$, forcing $a = c = 1$ and $b = \pm 1$, and a determinant-one basis change interchanges the two signs. Thus the oriented lattice is A_2 . The same lattice occurs for X by Theorem 5.1, so the Shioda–Inose classification [17, Theorem 4] gives $X_{0,1} \simeq X$. $\square$

The quotient has the unscaled lattice A_2 , while its Kummer companion has $A_2(2)$.

Proof of the Inose assertion in Theorem 6.1. Section 6.2 identifies the marked companion for every pair (A, B) with the product pair of Proposition 6.2, so its fixed-point complement is Oka, and Proposition 6.4 provides the equivariant diffeomorphism, biholomorphic near the fixed points, to the seed pair. Lemma 6.5 and Theorem 5.1 make the seed quotient Oka, after which Corollary 2.7 proves every $X_{A,B}$ Oka. $\square$

6.5 Every K3 surface of Picard rank twenty

The last part of Theorem 6.1 rests on the Shioda–Inose realization, not on an isogeny implication for the Oka property: that realization identifies every Picard-rank-twenty K3 surface with an elliptic model having two II^* fibres. We recall its scope and normalize the model, since the normalization is where the arithmetic of the parameters has to be checked.

Given an oriented positive even matrix

$$Q = \begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix}, \quad d = b^2 - 4ac < 0,$$

the surjectivity proof of Shioda–Inose [17, Theorem 4] uses the product of elliptic curves with period parameters

$$\tau_1 = \frac{-b + \sqrt{d}}{2a}, \quad \tau_2 = \frac{b + \sqrt{d}}{2}, \quad \Im \sqrt{d} > 0.$$

There is no condition $\gcd(a, b, c) = 1$. Their Theorem 3 gives the unscaled lattice and period compatibility for the resulting K3 surface, and their Lemma 3.1 gives a section and two II^* fibres in every case, the remaining fibres being 2I_2 , $\text{I}_2 + 2\text{I}_1$, 4I_1 , 2II , or IV . The orientation is preserved by the period compatibility, in the period-ratio convention specified in their §4. Therefore the injectivity of Theorem 4 identifies any given Picard-rank-twenty K3 with one of these two- II^* surfaces, and equal CM factors as well as the discriminants three and four are included.

Lemma 6.6. *Every complex elliptic K3 surface with a section and two II^* fibres is isomorphic to $X_{A,B}$ for finite $A, B \in \mathbb{C}$.*

Proof. Move the two fibre values to zero and infinity. The global minimal short Weierstrass form has $a_4 \in H^0(\mathbb{P}^1, \mathcal{O}(8))$ and $a_6 \in H^0(\mathbb{P}^1, \mathcal{O}(12))$, and the two II^* valuations force

$$a_4(t) = c_4 t^4, \quad a_6(t) = t^5(c_2 t^2 + c_1 t + c_0), \quad c_0 c_2 \neq 0.$$

Indeed, at a II^* endpoint the minimal a_6 order is exactly five, since an order at least six together with the a_4 order at least four would violate minimality; hence neither endpoint coefficient vanishes, while c_4 and c_1 may vanish.

We choose nonzero α, β with $\alpha^2 = c_0/c_2$, $\beta^6 = c_0\alpha^5$. The change $t = \alpha T$, $x = \beta^2 X$, $y = \beta^3 Y$ gives (26), with

$$A = -\frac{c_4\alpha^4}{3\beta^4}, \quad B = -\frac{c_1\alpha}{2c_0}.$$

Both are finite, and the base automorphism together with the constant Weierstrass scaling identifies the compact minimal models, including every singular component. $\square$

Proof of the Picard-rank-twenty assertion in Theorem 6.1. The original realization and classification identify the given compact K3 surface with a Jacobian elliptic K3 having two II^* fibres, and Lemma 6.6 then identifies it with a finite-parameter $X_{A,B}$. The first part of Theorem 6.1 proves that surface Oka, which completes the proof. $\square$

7 Two fibrations of small intersection

For the two arguments below, let $f, g : X \rightarrow \mathbb{P}^1$ be arbitrary distinct genus-one fibrations on a K3 surface, with fibre classes F, G , and let R be the reduced union of the irreducible curves contracted by both fibrations. The Hodge index theorem gives $F \cdot G > 0$, since equality would make the primitive nef isotropic classes equal; hence $H = F + G$ has positive square and is big and nef, so X is projective. The tensor product of the two globally generated bundles of the fibrations is globally generated, and the product map has a Stein factorization

$$X \xrightarrow{q} W \xrightarrow{\nu} \mathbb{P}^1 \times \mathbb{P}^1,$$

where q is birational with connected fibres, W is normal projective, and ν is finite of degree $F \cdot G$. The ample bundle $L_0 = \nu^* \mathcal{O}(1, 1)$ satisfies $q^* L_0 = H$. An irreducible curve C is contracted by both fibrations exactly when $F \cdot C = G \cdot C = 0$, and since ν is finite these are exactly the curves contracted by q ; the same observations show that q is an isomorphism off R . The big-nef contraction theorem for K3 surfaces [11, Chapter 2, Remark 3.4] identifies the connected exceptional configurations as ADE divisors and their images as rational double points. Every component of R is therefore a smooth rational (-2) -curve, and their intersection matrix is negative definite.

Proposition 7.1. *Let $f, g : X \rightarrow \mathbb{P}^1$ be distinct genus-one fibrations on a K3 surface, with fibre classes F, G satisfying $F \cdot G \leq 3$. If R is their reduced common contracted divisor, then $X \setminus R$ is Oka.*

Proof. There are two steps: the intersection bound excludes common critical points outside R , and the two fibrewise actions then move every point of the complement into a spanning locus. We begin with the first, which is the only place where the numerical hypothesis is used.

We apply the preceding setup to the two given fibrations. The finite map $\nu : W \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ has degree $d = F \cdot G \leq 3$, and the normal surface W is Cohen–Macaulay, so its finite map to the smooth surface $\mathbb{P}^1 \times \mathbb{P}^1$ is flat; moreover q is an isomorphism away from R . Suppose that $x \notin R$ were critical for both fibrations, write $A = \mathcal{O}_{X,x}$ with maximal ideal $\mathfrak{m}$, and let u, v be the local coordinate pullbacks, so that $(u, v) \subset \mathfrak{m}^2$. The first two pieces of the maximal-ideal filtration on $A/(u, v)$ have dimensions one and two, while the vector

space $\mathfrak{m}^2/\mathfrak{m}^3$ has dimension three and the initial quadratic terms of u, v span at most two dimensions; the third piece therefore has dimension at least one. It follows that

$$\text{length } A/(u, v) \geq 1 + 2 + 1 = 4,$$

contradicting the length $d \leq 3$ of the entire finite flat fibre. Hence there is no common critical point outside R .

Now we show that the spanning locus is reachable. The complete additive sprays of Lemma 3.1 preserve every irreducible component of a fibre, because a connected flow cannot permute a finite set of components; they therefore preserve R and its complement M . The product map has finite fibres on M . Let

$$U_0 = \{x \in M : df_x \text{ and } dg_x \text{ are independent}\}.$$

If $x \in M \setminus U_0$, the preceding step gives a nonzero differential, say df_x , and the corresponding flow gives a nonconstant small disk in the f -fibre. Were that disk to stay outside U_0 , then dg would vanish on its tangent line $\ker df$, so the disk would lie in a finite product fibre, which is impossible. Thus every point can be moved into U_0 , and Lemma 3.2 proves that M is Oka. $\square$

Remark 7.2. Let $L = \mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1}(1, 1)$ and choose smooth transverse branch curves defined by general sections P, Q of $L^{\otimes 2}$. The bidouble cover

$$\pi : Y \longrightarrow \mathbb{P}^1 \times \mathbb{P}^1, \quad u^2 = P, \quad v^2 = Q,$$

with u, v valued in L , is smooth. Adjunction gives $K_Y \simeq \mathcal{O}_Y$, and

$$\pi_* \mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-1, -1)^{\oplus 2} \oplus \mathcal{O}(-2, -2)$$

gives $H^1(Y, \mathcal{O}_Y) = 0$ by the Künneth formula. Hence Y is a K3 surface. On a general ruling the two quadratic covers have distinct nonempty branch sets, so their function-field extensions are linearly disjoint. Each projection therefore has connected general fibre. Its degree-four map to $\mathbb{P}^1$ has four simple branch values, each contributing two to the ramification divisor, and Riemann–Hurwitz gives genus one. The two fibre classes have intersection four.

Since π is finite, the common contracted divisor is empty. At each of the eight intersections of the two branch curves, local coordinates give $\pi(u, v) = (u^2, v^2)$, so the differentials of both fibrations vanish. Thus the no-common-critical-point conclusion in Proposition 7.1 fails at degree four. The surface is nevertheless Oka by the genus-one theorem proved below; common critical points obstruct this particular complement argument, not the Oka property.

7.1 Matched exceptional neighbourhoods

The geometric comparison below preserves both fibrations and the whole common exceptional neighbourhood while moving to Picard rank twenty. It applies to arbitrary distinct fibrations; the intersection bound enters only through Proposition 7.1.

Proposition 7.3. *Let X be a K3 surface with two distinct genus-one fibrations and fibre classes F, G . Let $R = \bigcup_{i=1}^k C_i$ be their reduced common contracted divisor. There is a small smooth marked deformation $\pi : \mathcal{X} \rightarrow B$ with central fibre X , such that:*

1. Both fibrations persist as genus-one fibrations on every fibre. The classes $F, G, [C_i]$ remain constant, and the C_i extend to labeled smooth relative rational curves with the same SNC intersection graph. Their union R_b is exactly the common contracted divisor on every fibre.
2. Some neighborhood V of R in X admits a holomorphic open embedding $j : V \times B \rightarrow \mathcal{X}$ over B , with $j_0 = \text{id}$ and $j(C_i \times B)$ equal to the corresponding relative curves in its image.
3. Every neighborhood of 0 in B contains a parameter b with $\rho(X_b) = 20$. For every such sufficiently small b there is a global diffeomorphism $h_b : X \rightarrow X_b$, inducing the marking and equal to j_b on a neighborhood of the whole R .

The base has dimension $18 - k$. If this dimension is zero, the central fibre itself is an allowed model.

Proof. We must produce, in one family, an ample class that survives the deformation, curves and fibrations that persist with them, an exceptional neighbourhood that is locally constant, and a nearby member of Picard rank twenty. The first step is to choose an ample class in the lattice generated by the two fibrations and the exceptional curves, so that preserving this lattice keeps the curves, the fibrations and the contracted locus unchanged; after that the lifted ADE components make the entire exceptional neighbourhood locally trivial, and a nearby rational period plane gives the Picard-rank-twenty model together with a matched global diffeomorphism.

We consider first the contraction and a preserved ample class. We apply the common setup to these two fibrations; recall that $q : X \rightarrow W$ contracts exactly R , that L_0 is ample with $q^*L_0 = H = F + G$, and that the components C_i form negative-definite ADE configurations.

For $k > 0$, the matrix $Q = (-C_i \cdot C_j)$ is block diagonal with finite ADE Cartan blocks, and each inverse block has positive entries: if A_Γ is the adjacency matrix, then

$$Q^{-1} = (2I - A_\Gamma)^{-1} = \frac{1}{2} \sum_{n \geq 0} (A_\Gamma/2)^n,$$

the spectral radius of A_Γ being below 2, while connectedness makes the relevant entries strictly positive. An integer multiple of $Q^{-1}\mathbf{1}$ therefore gives positive integral coefficients d_i such that the effective cycle $D = \sum_i d_i C_i$ has $D \cdot C_i < 0$ for every i .

The bundle $\mathcal{O}_X(-D)$ has positive degree on every irreducible component of every exceptional fibre, and hence is ample on that proper curve, including its nilpotent thickening, and it is also ample on zero-dimensional fibres. Since fibrewise ampleness for a proper morphism implies relative ampleness [19, Lemma 37.50.3], the class

$$P = NH - D \tag{46}$$

is ample for every sufficiently large integer N [19, Lemma 29.38.7(1)]. For $R = \emptyset$, the map q is an isomorphism and H is ample; take $D = 0$ and $P = H$.

Let L be the primitive saturation in $H^2(X, \mathbb{Z})$ of the span of $F, G, [C_1], \dots, [C_k]$. Over $\mathbb{Q}$ this span is the orthogonal sum of the hyperbolic plane spanned by F, G and the negative-definite exceptional lattice, so

$$\text{rank } L = k + 2, \quad \text{sign}(L) = (1, k + 1), \quad k \leq 18. \tag{47}$$

Saturation changes neither the rational span nor these period equations, and the actual ample class P lies in the original integral span.

We next put the curves and the fibrations into a period family. Local Torelli identifies a small part of the marked K3 deformation space with its period domain, and we restrict to a small coordinate ball in

$$\{[\omega] \in \mathbb{P}(L_{\mathbb{C}}^{\perp}) : (\omega, \omega) = 0, (\omega, \bar{\omega}) > 0\}. \quad (48)$$

Its dimension is $18 - k$. All classes in L stay of type $(1, 1)$, and their line bundles extend over a sufficiently small contractible Stein base by the relative exponential sequence, since their images in $R^2\pi_*\mathcal{O}$ vanish; see [11, Chapter 6, Section 2.4].

For a component C_i the central cohomology dimensions are

$$h^0(\mathcal{O}_X(C_i)) = 1, \quad h^1(\mathcal{O}_X(C_i)) = h^2(\mathcal{O}_X(C_i)) = 0.$$

Indeed, a second effective representative of its class would intersect C_i negatively and would contain it, and the remaining effective divisor would have zero Kähler area; Serre duality and Riemann–Roch give the two vanishings. Semicontinuity and base change extend the one-dimensional space of sections to the family, whose zero locus is a relative effective Cartier divisor. The central zero locus is smooth, so compactness and properness give smooth nearby fibres after shrinking, and they are rational curves. The fixed intersection numbers, openness of transverse intersection, and separation of the finitely many central crossing points preserve the complete labeled SNC graph.

The fibration classes satisfy

$$h^0(F) = h^0(G) = 2, \quad h^1(F) = h^2(F) = h^1(G) = h^2(G) = 0.$$

The latter assertions follow from non-effectivity of their negatives, duality, and Riemann–Roch, so base change gives rank-two bundles of sections. The cokernels of the relative evaluation maps have proper support disjoint from the central fibre, so both evaluations remain surjective after shrinking; trivializing the rank-two bundles gives actual relative maps to $\mathbb{P}^1$.

The fibre classes remain primitive, isotropic, and distinct. In the Stein factorization of either map, a nontrivial finite degree would make its class divisible by that degree, and its intermediate base has genus zero, because the irregularity of a K3 is zero. Thus the fibres are connected, and Bertini and adjunction show that the general fibres are smooth of genus one.

The bundle P remains ample: extend a positive Hermitian metric from the central fibre, then retain positivity of vertical curvature on all nearby compact fibres. The cycle $D_b = \sum_i d_i C_{i,b}$ remains effective, and if $C \subset X_b$ is any irreducible curve other than these labeled components, then

$$(F_b + G_b) \cdot C = \frac{P_b \cdot C + D_b \cdot C}{N} > 0. \quad (49)$$

Here $P_b \cdot C > 0$ by ampleness and $D_b \cdot C \geq 0$ by positivity of intersections of distinct curves, while every labeled component is still orthogonal to both fibrations. Formula (49) proves assertion (i), including at all Picard jumps.

What remains is to make the exceptional neighbourhood locally constant. For each connected component of R , take its resolution neighbourhood over a rational double point. Wahl [20, Sections 1.1–1.5] describes the local analytic minimally versal deformation space of such a resolution neighbourhood: it is the complex reflection representation of its

ADE Weyl group, and the locus on which a given simple exceptional component lifts is its simple-root reflection hyperplane, the lifting being a relative effective Cartier divisor. Since the deformation space here concerns the analytic resolution neighbourhood itself, its triviality will provide the required biholomorphic comparison.

Restriction of the family gives a classifying map of analytic germs

$$\lambda : (B, 0) \longrightarrow (\text{Res}, 0).$$

Each actual relative curve constructed above is a lifting of the corresponding simple exceptional component, so λ lands in all the simple-root hyperplanes. The simple roots form a basis, so their hyperplanes intersect only at 0; consequently $\lambda = 0$, and pulling back the analytic versal family along this constant map gives a product. This yields an isomorphism of the germs of families along that entire connected exceptional configuration.

There are finitely many configurations, so we may take disjoint neighbourhoods for them and shrink the base once. An isomorphism of germs along a compact set has representatives on actual open neighbourhoods, and a finite product-neighbourhood cover of $R \times \{0\}$ provides a common domain $V \times B$; the inverse germ makes the representative an open embedding, and composing with the inverse of its central restriction gives $j_0 = \text{id}$. The relative curves in this product agree with the labeled curves by uniqueness of their sections, which proves (ii). It also gives a simultaneous contraction: use the original local ADE contraction times B on these product neighbourhoods and the identity off the exceptional set, the restrictions agreeing where the local contractions are isomorphisms.

Finally we arrange the Picard rank and the global matched diffeomorphism. The rational space $L_{\mathbb{Q}}^{\perp}$ has real signature $(2, 18 - k)$, and rational oriented positive two-planes are dense among all its oriented positive real two-planes, by rational approximation of an oriented basis and openness of positivity. Each such rational plane has an integral orthogonal complement of rank twenty in the K3 lattice, so its period has Picard number twenty, and local Torelli realizes sufficiently nearby planes in the same base (48), where all the preceding properties hold.

For a chosen parameter b , take a smooth path from 0 to b in this small base. A smooth horizontal lift for the proper smooth family along the path can be changed by a vertical vector field and a cutoff so as to agree with the product horizontal lift on a smaller neighbourhood of the whole R : explicitly, if V_0 is any lift and V_{prod} the product lift, then

$$V_0 + \eta(V_{\text{prod}} - V_0)$$

is a lift, where η is one on a smaller product neighbourhood of R and supported in the given product chart. Its flow exists over the entire compact path, and a point in the smaller neighbourhood follows exactly its product trajectory, so the final diffeomorphism equals j_b there; as an Ehresmann transport it induces the marking. This proves (iii).

If $k = 18$, the preserved lattice already has rank twenty, so the central fibre is itself an admissible model and the base can be a point; if R is empty, the neighbourhood condition is vacuous and the same period argument applies. $\square$

Corollary 7.4. *A K3 surface carrying two distinct genus-one fibrations whose fibre classes have intersection at most three is Oka.*

Proof. Let R be the common contracted divisor. Proposition 7.1 makes $X \setminus R$ Oka, and Proposition 7.3 provides a Picard-rank-twenty model Y with the same two classes and a biholomorphic comparison near the whole exceptional divisor. Its complement is Oka by the same small-intersection proposition, and Y is Oka by Theorem 6.1, so Theorem 2.1 proves the assertion. When R is empty, the complement proposition already suffices. $\square$

8 An intersection-two model for every Jacobian

The remaining ingredient is a supply of surfaces to which Corollary 7.4 applies and whose Jacobians exhaust all possibilities; the next theorem provides it.

Theorem 8.1. *For every complex Jacobian elliptic K3 surface J , there is a projective K3 surface Y with two primitive nef isotropic classes f, D satisfying*

$$f \cdot D = 2, \quad \operatorname{div}_{\operatorname{NS}(Y)}(f) = 2,$$

such that the underlying K3 surface of the Jacobian of (Y, f) is isomorphic to J .

Proof. The strategy is to change the period so that the new surface has a genus-one fibration of multisection index two together with a second fibration meeting it twice; the degree-two relative Picard surface then recovers the original period by an integral Mukai-lattice calculation, and Global Torelli identifies its underlying K3 surface with J .

We begin with the period and the divisibility. Mark the K3 lattice using the original fibre and zero section of J :

$$H^2(J, \mathbb{Z}) = \Lambda = U \langle f, e \rangle \oplus \Lambda', \quad e = [O] + f, \quad \Lambda' \simeq U^2 \oplus E_8(-1)^2.$$

We write $\operatorname{NS}(J) = U \oplus W$, where W is negative definite, and let $\sigma \in \Lambda'_{\mathbb{C}}$ be its period; the primitive nonzero transcendental lattice T_J lies in Λ' .

There is a primitive integral isotropic $w \in \Lambda'$ pairing oddly with some $t \in T_J$. Indeed, integral isotropic vectors generate Λ' : if a, b span one hyperbolic summand and z lies in its even orthogonal complement, then

$$z + a - \frac{z^2}{2}b$$

is isotropic, and these vectors together with a, b generate the lattice. Were all isotropic vectors to pair evenly with T_J , unimodularity would imply $T_J \subset 2\Lambda'$, contrary to primitiveness; moreover, dividing a chosen vector by its common divisor preserves an odd pairing and makes it primitive.

We set

$$\sigma' = \sigma - \frac{(w, \sigma)}{2}f, \quad D = 2e + w. \quad (50)$$

Then $(\sigma')^2 = 0$ and $(\sigma', \overline{\sigma'}) = (\sigma, \overline{\sigma}) > 0$, and the classes f, D are integral Hodge classes with Gram matrix $U(2)$. Period-map surjectivity gives a marked K3 surface with this period, and it is projective because $(f + D)^2 = 4$.

For an integral Hodge class $x = ae + bf + v$, with $v \in \Lambda'$, the Hodge equation gives $2v - aw \in W$, and pairing with the chosen t forces a even. Since $(f, x) = a$ and $(f, D) = 2$, the divisibility of f in the Néron–Severi lattice is exactly two.

We next look for a chamber in which both fibrations are nef. No (-2) -root C can pair with f, D with opposite strict signs: after changing its sign, such a root would satisfy

$$(f, C) = 2a > 0, \quad (D, C) = -b < 0$$

for positive integers a, b , its projection to their hyperbolic plane would be $-(b/2)f + aD$, of square $-2ab$, and the orthogonal complement is negative definite. Thus $C^2 = -2$ forces $a = b = 1$ and zero orthogonal part, giving $C = -f/2 + D$, which is impossible because f is primitive in Λ .

We choose the positive-cone sign and use root reflections to make $f + D$ big and nef; the preceding observation then makes both f, D nef in that chamber. This is an inverse re-marking by algebraic-root reflections, which fix the period, so formulas (50) remain valid in the chosen marking. The class D is primitive: any proper division would, since $(f, D) = 2$, give a class of fibre degree one, contradicting the divisibility just proved. Note that primitive nef isotropic classes give connected genus-one fibrations, and that the first of them has multisection index two.

It remains to recover the Jacobian. Let $J(Y)$ be the Jacobian of the fibration defined by f . On each smooth fibre, tensoring by $\mathcal{O}_Y(D)$ identifies its degree-zero and degree-two Picard curves, so uniqueness of the relatively minimal smooth compactification gives $J^2(Y) \simeq J(Y)$; see [11, Chapter 11, Remark 4.4]. We choose a generic ample polarization for the primitive Mukai vector $(0, f, 2)$. The corresponding smooth moduli K3 is the compact relative degree-two Picard surface by [11, Chapter 11, Section 4.2 and Proposition 4.5], the identification being made over the smooth-fibre locus and extended by minimal compactification.

The integral Mukai Hodge isometry now gives

$$H^2(J(Y), \mathbb{Z}) \simeq (0, f, 2)^\perp / \mathbb{Z}(0, f, 2) \quad (51)$$

[11, Chapter 11, Corollary 4.7 and equation (4.4)], and the isometry applies also to the possibly non-fine moduli space [11, Chapter 10, Remark 3.7; Chapter 11, Remark 4.9].

In this quotient put

$$u = (0, 0, 1), \quad e_J = (1, 2e, 0), \quad f_J = -u.$$

The last two vectors span a hyperbolic plane orthogonal to $(0, \Lambda', 0)$. Together with that lattice they generate the quotient: every perpendicular vector is $(r, 2re + bf + c, s)$ and becomes

$$re_J + c + (s - 2b)u.$$

The period in this decomposition is $\sigma - (w, \sigma)f_J$. The integral Eichler isometry

$$f_J \mapsto f_J, \quad c \mapsto c - (c, w)f_J \quad (c \in \Lambda'), \quad e_J \mapsto e_J + w$$

preserves the pairing because $w^2 = 0$, and it takes the original period σ to this one. Thus $J(Y)$ and J have integrally Hodge-isometric second cohomology. We choose the positive-cone sign and apply algebraic-root reflections to align their ample chambers; Global Torelli then gives an isomorphism of the underlying K3 surfaces. $\square$

We can now dispose of the Jacobians. Let J be any complex Jacobian elliptic K3 surface. Theorem 8.1 provides a projective K3 surface Y with two fibrations of fibre classes f, D satisfying $f \cdot D = 2$ and $J(Y) \simeq J$ as underlying compact surfaces, and Corollary 7.4 makes Y Oka. Theorem 4.2, applied to the fibration (Y, f) , makes its Jacobian $J(Y)$ Oka, and the surface isomorphism then makes J Oka. Thus

$$Y \text{ Oka} \implies J(Y) \text{ Oka} \implies J \text{ Oka}.$$

Proof of Theorem 1.1. Let X be an analytic genus-one K3 surface. Its Jacobian J is a projective Jacobian K3 surface, and X is an analytic torsor of it [11, Chapter 11, Sections 4.3 and 5.3]. The preceding argument makes J Oka, and Theorem 4.2, now applied to X , makes X Oka. $\square$

Corollary 8.2. *Every complex K3 surface with two distinct genus-one fibrations is Oka.*

This follows from Theorem 1.1.

9 Finite symplectic quotients of two-tori

A *generalized Kummer surface* is a K3 surface obtained as the minimal resolution of A/G , where A is a complex two-torus and G is a finite group of holomorphic automorphisms preserving a nonzero holomorphic two-form. Translations are allowed in G , and the term is used here only for surfaces; the case $G = \{1, -1\}$ gives the ordinary Kummer surfaces.

The strategy is the same as before: we construct an equivariant projective model, compare the resolved pairs, and only then invoke an Oka result, the comparison surface being provided by the genus-one theorem, Theorem 1.1.

Proposition 9.1. *Let $X \rightarrow A/G$ be a generalized Kummer surface, and let E be its reduced exceptional divisor. There are a projective generalized Kummer surface $Y \rightarrow B/G$, with reduced exceptional divisor F , and a smooth diffeomorphism $h : X \rightarrow Y$. It is biholomorphic near E and maps its components to those of F . The group acts on A and B by the same affine diffeomorphisms of one real torus. Moreover, $\rho(Y) \geq 17$, so Y admits a genus-one fibration.*

9.1 Equivariant deformation and local comparison

We write $A = (V/\Lambda, I_0)$, where V is a real vector space of dimension four, Λ is a lattice and I_0 is a linear complex structure. Every holomorphic automorphism of a torus is affine, and its linear part preserves Λ and commutes with I_0 ; this is what allows us to deform the complex structure while keeping the group action fixed.

Lemma 9.2. *There is a smooth path I_t , $0 \leq t \leq 1$, of linear complex structures commuting with the linear parts of G , beginning at I_0 , such that $(V/\Lambda, I_1)$ is projective. The same affine action of G preserves a holomorphic two-form for every I_t .*

Proof. Let Γ be the finite group of linear parts. Average an I_0 -Hermitian inner product to obtain a Γ -invariant one, denoted by g , and put $\omega_0(u, v) = g(I_0 u, v)$. The vector space of Γ -invariant alternating forms is defined over $\mathbb{Q}$ relative to Λ , since its defining equations have integral coefficients, so its rational points are dense. We choose a rational invariant alternating form ω_1 sufficiently close to ω_0 and put $\omega_t = (1 - t)\omega_0 + t\omega_1$.

Define the skew-adjoint endomorphism B_t by $g(B_t u, v) = \omega_t(u, v)$; for a sufficiently close choice of ω_1 , every B_t is invertible. We set

$$I_t = B_t(-B_t^2)^{-1/2}, \quad (52)$$

where the square root is the positive self-adjoint one. These complex structures commute with Γ , depend smoothly on t , and begin at I_0 . The endomorphism B_t commutes with its positive factor, so ω_t has type $(1, 1)$ for I_t and

$$\omega_t(u, I_t u) = g((-B_t^2)^{1/2} u, u) > 0 \quad (u \neq 0).$$

Thus ω_t is a translation-invariant Kähler form, and an integral multiple of ω_1 is integral on Λ ; the Riemann bilinear criterion therefore makes $(V/\Lambda, I_1)$ an abelian surface.

It remains to check that the group still preserves holomorphic forms. For each $\gamma \in \Gamma$, its complex determinant relative to I_t varies continuously and belongs to the finite set of roots of unity whose orders divide the order of γ ; it equals one at $t = 0$ and hence throughout the path. Since translations act trivially on holomorphic forms, every affine element of G preserves the holomorphic volume form for I_t . $\square$

Lemma 9.3. *Let $P \subset V/\Lambda$ be the set of points with nontrivial stabilizer. There is a G -equivariant smooth diffeomorphism*

$$(V/\Lambda, I_0) \longrightarrow (V/\Lambda, I_1)$$

which fixes P pointwise and is biholomorphic near P .

Proof. The set P is finite. Indeed, a nonidentity linear part is a finite-order complex matrix of determinant one; if it had an eigenvalue one, both eigenvalues would be one, and semisimplicity would make it the identity. Thus $1 - \gamma$ is invertible for every nonidentity linear part γ , an affine automorphism with that linear part has finitely many fixed points, and a nontrivial translation has no fixed point at all.

Let $L_t \in \mathrm{GL}(V)$ solve

$$\dot{L}_t = \frac{1}{2} I_t \dot{I}_t L_t, \quad L_0 = \mathrm{id}. \quad (53)$$

Differentiating $I_t^2 = -\mathrm{id}$ gives $[\frac{1}{2} I_t \dot{I}_t, I_t] = \dot{I}_t$, whence $I_t L_t = L_t I_0$. The coefficient in (53) commutes with Γ , so L_t does as well. In particular, L_1 is a complex-linear isomorphism from (V, I_0) to (V, I_1) , joined to the identity through real invertible maps commuting with every stabilizer.

We choose disjoint small coordinate balls centred at the points of P , using a Γ -invariant norm and the same radius on each G -orbit. At an orbit representative, the stabilizer acts linearly in centred coordinates and commutes with every L_t , so Lemma 2.8, applied to this stabilizer and the isotopy L_t , gives an equivariant diffeomorphism supported in the ball and equal to L_1 near its centre.

Transport this map to the other balls by the affine action. Stabilizer equivariance makes the construction well-defined, and commutation of L_1 with Γ makes every transported map equal to L_1 in centred coordinates near its centre. Gluing these maps with the identity outside the balls gives a smooth diffeomorphism which commutes with G , fixes P pointwise, and is biholomorphic near P . $\square$

9.2 Resolution and the Picard number

Lemma 9.4. *Let $X \rightarrow A/G$ be a generalized Kummer surface, and let r be the number of irreducible exceptional curves. Then $r \geq 16$. If X is projective, then $\rho(X) \geq 17$ and X admits a genus-one fibration.*

Proof. At a point with nontrivial stabilizer the local group is a finite subgroup of $\mathrm{SL}_2(\mathbb{C})$, so the quotient has a rational double point, and the exceptional curves form an ADE configuration of smooth rational curves with negative-definite intersection matrix; see [11, Chapter 14, Section 0.3].

We write E for the exceptional divisor and $U = X \setminus E$. Every connected component of E is a tree of rational curves, hence $H_1(E, \mathbb{R}) = 0$ and $H_2(E, \mathbb{R}) \simeq \mathbb{R}^r$. Duality for the pair (X, U) and its cohomology sequence give

$$H_2(E, \mathbb{R}) \longrightarrow H^2(X, \mathbb{R}) \longrightarrow H^2(U, \mathbb{R}) \longrightarrow H_1(E, \mathbb{R}).$$

The first map sends component classes to their Poincaré duals and is injective by negative definiteness, so

$$\dim H^2(U, \mathbb{R}) = 22 - r. \quad (54)$$

Let $P \subset A$ be the finite set of points with nontrivial stabilizer. Then $A \setminus P \rightarrow U$ is an unramified finite covering with group G , and transfer and averaging identify $H^2(U, \mathbb{R})$ with

$H^2(A \setminus P, \mathbb{R})^G$. Removing finitely many points from a real four-manifold does not change its second cohomology, so

$$H^2(U, \mathbb{R}) \simeq H^2(A, \mathbb{R})^G, \quad \dim H^2(U, \mathbb{R}) \leq 6.$$

Equation (54) proves $r \geq 16$. In the projective case an ample class is independent of the negative-definite exceptional span, giving $\rho(X) \geq r + 1 \geq 17$, and a projective K3 surface of Picard rank at least five has a genus-one fibration by [11, Chapter 11, Proposition 1.3]. $\square$

Proof of Proposition 9.1. Lemmas 9.2 and 9.3 give an abelian surface B with the same affine action and an equivariant diffeomorphism $A \rightarrow B$, biholomorphic near the nonfree set; the nonfree sets are the same subsets of the real torus. The induced map $A/G \rightarrow B/G$ is smooth on the free loci and biholomorphic near each quotient singularity, so by uniqueness of the minimal resolution of a normal surface singularity the biholomorphic germ comparisons lift to the minimal resolutions. They agree with the map on the smooth loci, yielding a smooth diffeomorphism $h : X \rightarrow Y$ biholomorphic near the full exceptional divisors.

The finite quotient of the abelian surface B is projective, and so is its minimal resolution Y . The invariant holomorphic two-form extends without zeros across the rational double point resolutions, and Y is diffeomorphic to the simply connected K3 surface X ; hence Y is a K3 surface. Lemma 9.4 gives its Picard-rank bound and its genus-one fibration. $\square$

9.3 Oka complements and the generalized Kummer theorem

Lemma 9.5. *For every generalized Kummer surface $X \rightarrow A/G$ with reduced exceptional divisor E , the complement $X \setminus E$ is Oka.*

Proof. For the finite nonfree set $P \subset A$, one has $X \setminus E = (A \setminus P)/G$ and the action on $A \setminus P$ is free. The complement of any finite set in a complex torus of dimension at least two is Oka by [5, Proposition 6.5], and the Oka property descends through an unramified covering by [6, Theorem 1.4]. $\square$

Proof of Theorem 1.2. Proposition 9.1 provides a projective genus-one K3 surface Y and a diffeomorphism of the resolved pairs, biholomorphic near their exceptional divisors. Those divisors are simple normal crossing divisors, and their complements are Oka by Lemma 9.5; Theorem 1.1 makes Y Oka, after which the transfer theorem, Theorem 2.1, makes X Oka. $\square$

Proof of Corollary 1.3. Let A be a complex two-torus. Inversion preserves its holomorphic two-form, and the minimal resolution of $A/\{\pm 1\}$ is the Kummer K3 surface $\text{Kum}(A)$, so Theorem 1.2 applies. $\square$

Corollary 9.6. *Every order-three generalized Kummer surface of Picard number eighteen is Oka, is nonprojective, has no genus-one fibration, and is not an ordinary Kummer surface. Such surfaces exist.*

Proof. The Picard-number-eighteen members occur in the order-three moduli description of [15, Introduction and Sections 2.6 and 5.3]. Their exceptional divisor consists of nine disjoint A_2 configurations, and the eighteen component classes span $\text{NS}(X) \otimes \mathbb{R}$, a negative definite space; so there is no polarization, and no nonzero square-zero class either, while a genus-one fibration would provide the second. Theorem 1.2 gives the Oka property.

It remains to see that there are no other irreducible curves. If C were distinct from the exceptional components E_i , then $C \cdot E_i \geq 0$; write $[C] = \sum_i a_i [E_i]$ and let M be their intersection matrix, so that $(a_i) = M^{-1}(C \cdot E_i)$. On each A_2 block,

$$M^{-1} = -\frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

so all $a_i \leq 0$, and pairing with a Kähler class contradicts the positive area of C . Finally, among nine A_2 pairs at most nine curves are mutually disjoint, while an ordinary Kummer surface contains sixteen disjoint smooth rational curves; thus X is not an ordinary Kummer surface. $\square$

10 Enriques surfaces

Let S be a smooth complex Enriques surface. Its canonical bundle has order two, and the associated canonical cover

$$\pi : X = \operatorname{Spec}_S(\mathcal{O}_S \oplus K_S^{-1}) \longrightarrow S$$

is a connected finite étale cover of degree two. The total space X is a smooth K3 surface and the covering involution ι has no fixed points. These classical facts hold for nodal and unnodal Enriques surfaces alike; see [11, Chapter 15, Section 4.3(ii)] and [3, Section 1.3].

Lemma 10.1. *The K3 cover X is projective, has $\rho(X) \geq 10$, and admits a genus-one fibration.*

Proof. The involution acts by -1 on $H^{2,0}(X)$: an invariant holomorphic two-form would descend to S , whereas $p_g(S) = 0$. The invariant integral lattice is

$$H^2(X, \mathbb{Z})^\iota \simeq U(2) \oplus E_8(-2)$$

[11, Chapter 14, Section 0.3(vii)]. It is of type $(1, 1)$ and lies in $\operatorname{NS}(X)$, so $\rho(X) \geq 10$, and the positive-square classes in $U(2)$ imply projectivity by the K3 projectivity criterion. A projective K3 surface of Picard rank at least five admits a genus-one fibration by [11, Chapter 11, Proposition 1.3]. $\square$

Proof of Corollary 1.4. Lemma 10.1 places the canonical K3 cover in Theorem 1.1, so X is Oka, and since $\pi : X \rightarrow S$ is an unramified holomorphic covering, the Oka property descends to S by [6, Theorem 1.4]. $\square$

A (-2) -curve on S is a smooth rational curve of self-intersection -2 . Its inverse image in X consists of two disjoint copies, since an unramified cover of $\mathbb{P}^1$ splits; in particular, the canonical cover is étale along the curves of a nodal Enriques surface.

Xie–Zhao prove that every unnodal Enriques surface is Oka in Theorem C and Corollary 3.5 of arXiv:2608.24392v2 [21], using an étale double cover by a Wehler K3 surface, and their introduction leaves the nodal case open. The canonical-cover argument above extends the conclusion to all smooth Enriques surfaces.

11 Oka loci in non-isotrivial K3 families

The last theorem is proved by a density argument on the period domain: integral isotropic hyperplanes meet every nonconstant period curve densely, and their intersections give genus-one fibres, to which Theorem 1.1 applies. This proves the density assertion in Theorem 1.5.

We write Λ for the K3 lattice, of signature $(3, 19)$, and write its period domain as

$$\Omega = \{[\sigma] \in \mathbb{P}(\Lambda_{\mathbb{C}}) : (\sigma, \sigma) = 0, (\sigma, \bar{\sigma}) > 0\}.$$

The corresponding oriented positive real plane is $P = \langle \operatorname{Re} \sigma, \operatorname{Im} \sigma \rangle$.

Lemma 11.1. *Let $\gamma : \Delta \rightarrow \Omega$ be a nonconstant holomorphic map. The points $t \in \Delta$ for which*

$$(\sigma(t), v) = 0 \quad \text{for some } 0 \neq v \in \Lambda, \quad (v, v) = 0,$$

are dense in Δ . Here σ is any local holomorphic lift of γ .

Proof. It suffices to work near a point where $\gamma' \neq 0$, since these points are dense in the disk. Move that point to 0 and choose a holomorphic lift $\sigma(t)$, and put $P_0 = \langle \operatorname{Re} \sigma(0), \operatorname{Im} \sigma(0) \rangle$ and $W = P_0^\perp \subset \Lambda_{\mathbb{R}}$; the restriction of the form to W has signature $(1, 19)$.

The projection ξ of $\sigma'(0)$ to $W_{\mathbb{C}}$ is nonzero. Indeed, differentiation of $(\sigma, \sigma) = 0$ gives $(\sigma'(0), \sigma(0)) = 0$, so if the projection vanished, $\sigma'(0)$ would be a multiple of $\sigma(0)$, contradicting $\gamma'(0) \neq 0$. Real null vectors span the Lorentzian space W , so there is a nonzero real vector $v \in W$ with $(v, v) = 0$ and $(\sigma'(0), v) \neq 0$; the holomorphic function

$$t \mapsto (\sigma(t), v)$$

therefore has a simple zero at 0.

It remains to replace v by an integral vector. Rational null vectors are dense in the real null cone of $\Lambda_{\mathbb{R}}$: for completeness, choose a rational hyperbolic plane with isotropic basis e, f satisfying $(e, f) = 1$, so that the affine chart of the null cone on which $(v, f) \neq 0$, after normalizing this pairing to one, is parametrized by

$$u \mapsto e + u - \frac{(u, u)}{2} f, \quad u \in \langle e, f \rangle^\perp.$$

Rational parameters are dense, this chart is dense in the real null cone, and real scalar multiples may likewise be approximated by rational multiples, so we may choose nonzero rational null vectors $v_j \rightarrow v$.

On a sufficiently small circle about 0, the function $(\sigma(t), v)$ has no zero and has exactly one zero inside. The functions $(\sigma(t), v_j)$ converge uniformly there, so Rouché's theorem gives zeros tending to 0, and multiplying each v_j by an integer makes it an integral isotropic vector without changing its zero set. Since the circle can be taken inside any prescribed neighbourhood of 0, this proves the density assertion. $\square$

Lemma 11.2. *A complex K3 surface with a nonzero integral isotropic Néron–Severi class admits a genus-one fibration.*

Proof. For a projective K3 surface this is the isotropic-class criterion, obtained by choosing a primitive class, its effective sign, and root reflections; see [11, Chapter 11, Proposition 1.3].

In the nonprojective case, let L be a nontrivial line bundle with $c_1(L)^2 = 0$. Riemann–Roch and Serre duality give

$$h^0(L) + h^0(L^{-1}) \geq \chi(L) = 2.$$

Both signs cannot be effective, since integration against a Kähler form is strictly positive on every nonzero effective divisor, so after changing the sign we have $h^0(L) \geq 2$. Remove the fixed divisor of its complete linear system and denote the moving class by M . Two general members have no common component, so $M^2 \geq 0$, and positive square would imply projectivity of the K3 surface; hence $M^2 = 0$.

Two general moving members now have intersection zero, so there are no base points, and the associated map has one-dimensional image, since a two-dimensional image would give positive self-intersection. In its Stein factorization $X \rightarrow B$ the base is a smooth compact curve and $H^1(B, \mathcal{O}_B)$ injects into $H^1(X, \mathcal{O}_X) = 0$, so $B \simeq \mathbb{P}^1$. A general connected fibre is smooth and has square zero, and adjunction gives genus one. $\square$

Proof of Theorem 1.5. Let $U \subset B$ be any nonempty open set. On small simply connected coordinate neighbourhoods the family has a marking and a holomorphic period map, and non-isotriviality, local Torelli and analytic continuation imply that this period map is not constant on any nonempty base open set. Thus inside U there is a small holomorphic disk on which it has a nonzero derivative at some point.

Lemma 11.1 provides fibres arbitrarily close to that point with a nonzero integral isotropic Néron–Severi class, and by Lemma 11.2 these fibres admit genus-one fibrations. Theorem 1.1 makes them Oka, so every nonempty base open set contains an Oka fibre; finally, Lárusson’s theorem [14, Corollary 8] makes the Oka locus of the proper holomorphic submersion a G_δ subset of B . $\square$

A Comparing arbitrary Kummer models

We close with the geometric comparison between Kummer models that the introduction refers to. For a complex two-torus A , write $Y_A = \mathrm{Bl}_{A[2]} A$ and $X_A = \mathrm{Kum}(A) = Y_A / \{\pm 1\}$. The fixed divisor $R_A \subset Y_A$ consists of sixteen copies of $\mathbb{P}^1$ with normal bundle $\mathcal{O}(-1)$, and its image in X_A has normal bundle $\mathcal{O}(-2)$, as the local squaring map in Corollary 2.7 shows.

Lemma A.1. *For any complex two-tori A and B , there is a smooth equivariant diffeomorphism $h : Y_A \rightarrow Y_B$, biholomorphic on a neighborhood of R_A .*

Proof. We choose an orientation-preserving isomorphism of the underlying real tori induced by an isomorphism of their lattices; it commutes with inversion and identifies the two-torsion points. In flat coordinates near corresponding two-torsion points, its local expression is an orientation-preserving real-linear isomorphism T .

Let L be a complex-linear isomorphism between the two complex tangent spaces. Both L and T preserve orientation, so $M = LT^{-1}$ lies in $\mathrm{GL}^+(4, \mathbb{R})$ and admits a smooth path M_t from the identity. We apply Lemma 2.8 with the inversion group $\{\pm \mathrm{id}\}$ in a coordinate ball of B centred at the corresponding two-torsion point; composing the resulting local diffeomorphism with the original torus isomorphism changes its expression from T to L near the centre.

We apply this construction in disjoint balls around the sixteen points. The resulting diffeomorphism $A \rightarrow B$ commutes with inversion and equals a complex-linear isomorphism near every blowup centre, and such an isomorphism lifts holomorphically to the blowup by its induced action on complex tangent lines. Off the exceptional divisors the local lifts agree with the map defined through the blowdown isomorphisms, so they glue to a smooth equivariant diffeomorphism h that is biholomorphic near R_A . $\square$

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